Contents

Chapter 1

Introduction to Data Visualization

It is difficult for the human mind to look at a list of numbers and identify the patterns in them, so we often use these numbers to make a picture. These pictures are called *graphs*, *charts*, or *plots*. Often, the right picture can make the meaning in the data obvious. **Data visualization** is the process of making pictures from numbers.

1.1 Types of Variables

Depending on the type of data and what you are trying to demonstrate about it, you will use different types of data visualizations. Before we get to the more complex types of data visualizations, let's look at the two types of variables we are usually looking to visualize: *categorical* and *numerical*.

Categorical or qualitative variables are those that can be put into categories. For example, the type of shoes a person wears (sneakers, sandals, etc.) is a categorical variable. The type of shoes sold is a categorical variable. The name of the salesperson (John, Jane, etc.) is also a categorical variable.

Numerical or quantitative variables are those that can be measured with numbers. For example, the number of boxes of cookies sold is a quantitative variable. The average temperature on a given day is also a quantitative variable. Quantitative variables can be further divided into *discrete* and *continuous* variables. Discrete variables are those that can only take on specific values, such as the number of pencils sold (you can't sell 3.5 pencils). Continuous variables can take on any value within a range, such as the average height of students in a class (which can be 5.5 feet, 1.65 m, etc.).

Further, we can also classify data into univariate, bivariate, and multivariate data. Univariate data has only one variable, such as the number of boxes of cookies sold. Bivariate data has two variables, such as the average temperature and total sales for a lemonade stand. Multivariate data has more than two variables, such as the average temperature, total sales, and type of ookie sold. We will only look at univariate and bivariate data!

1.2 How do we describe data?

There are different descriptive methods for representing data, and they can usually be grouped into three categories: *tabular*, *graphical*, and *numerical*.

Tabular methods include tables, which are a way of organizing data into rows and columns. They are useful for displaying raw data and making it easy to look up specific values. However, they can be difficult to interpret when dealing with large datasets or when trying to identify patterns.

Graphical methods include bar charts, line graphs, pie charts, and scatter plots. They are useful for visualizing data and making it easier to identify patterns and trends.

Numerical methods include measures of central tendency (mean, median, mode) and measures of variability (range, variance, standard deviation). They are useful for summarizing data and providing a numerical representation of the data's characteristics.

We will focus on graphical methods in this chapter, but we will also touch on tabular and numerical methods in later chapters.

1.3 Univariate Data

We will start with univariate data, which has only one variable. For example, the number of boxes of cookies sold by each salesperson is univariate data. Bar charts, line graphs, and pie charts are all good for visualizing univariate, *categorical* data. We will explore these types of graphs in the next sections.

1.3.1 Bar Chart

Bar charts are typically used to show the values of a categorical variable. Each bar represents a category, and the height of the bar represents the value of that category. Bar charts can have vertical or horizontal bars to represent this data.

Here is an example of a bar chart.

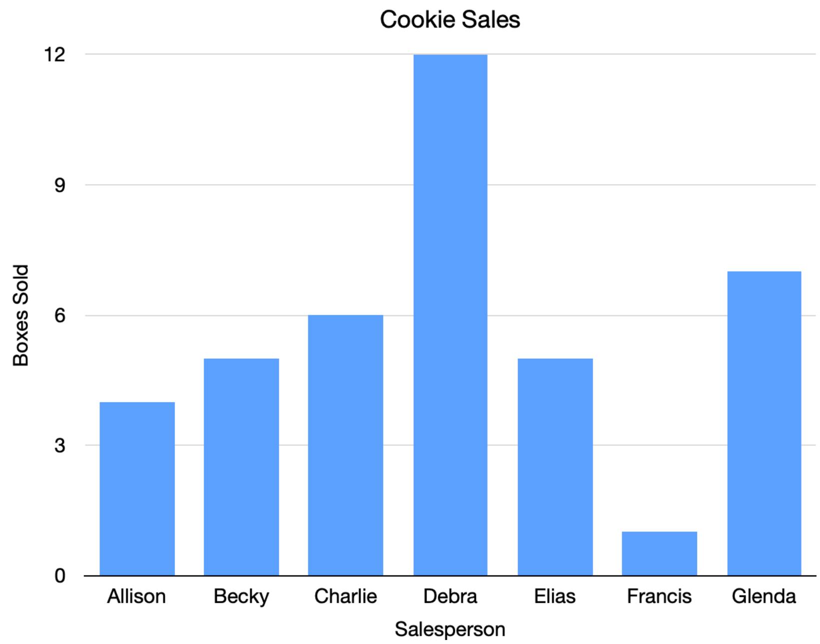


Figure 1.1: A bar chart showing cookie sales per person.

Each bar represents the cookie sales of one person. For example, Charlie has sold 6 boxes of cookies, so the bar goes over Charlie's name and reaches to the number 6.

Looking at this chart, you probably think, "Wow, Debra has sold a lot more cookies than anyone else, and Francis has sold a lot fewer."

The same data could be in a table like this:

Salesperson	Boxes Sold
Allison	4
Becky	5
Charlie	6
Debra	12
Elias	5
Francis	1
Glenda	7

A table (especially a large table) is often just a bunch of numbers. A chart helps our brains understand what the numbers mean.

Bar charts can also go horizontally.

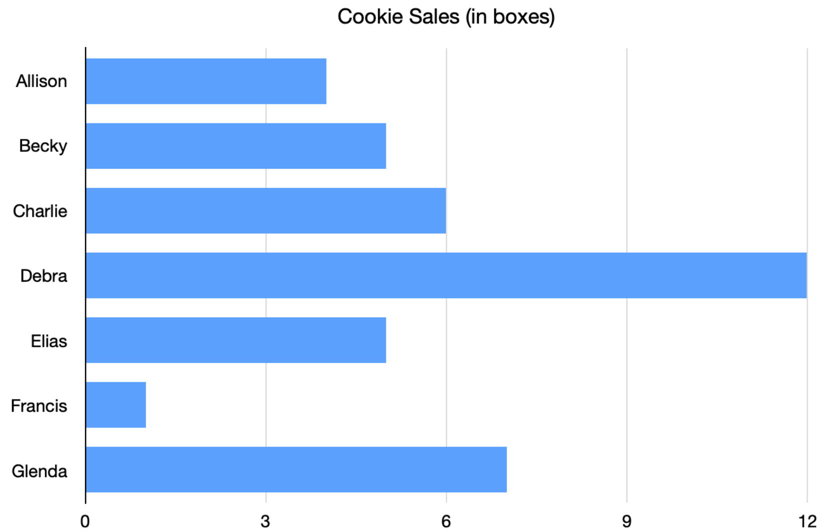


Figure 1.2: A horizontal bar chart showing the same cookie sales data.

Sometimes we use colors to explain what contributed to the number.

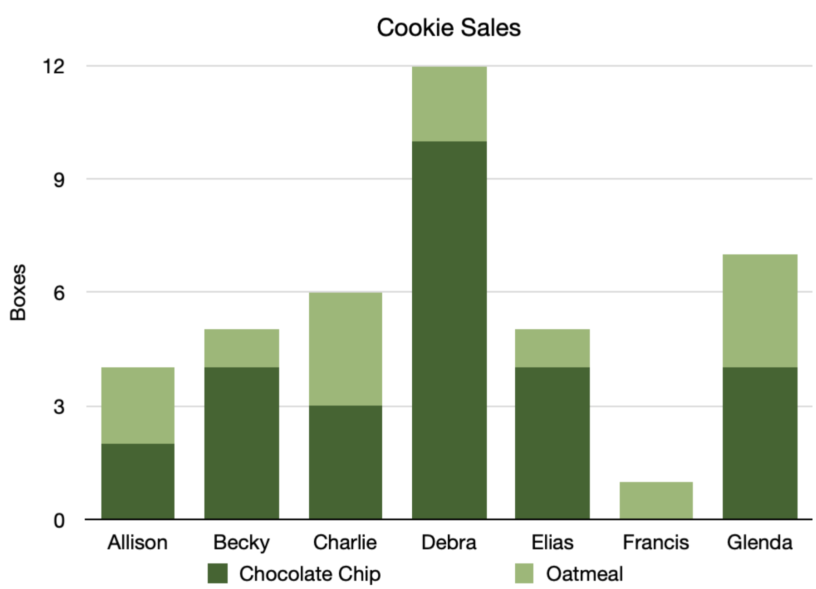


Figure 1.3: A bar chart with different colors showing types of cookies.

This tells us that Becky sold more boxes of chocolate chip cookies than boxes of oatmeal cookies.

Make Bar Graph

Go back to your compound interest spreadsheet and make a bar graph that shows both balances over time:

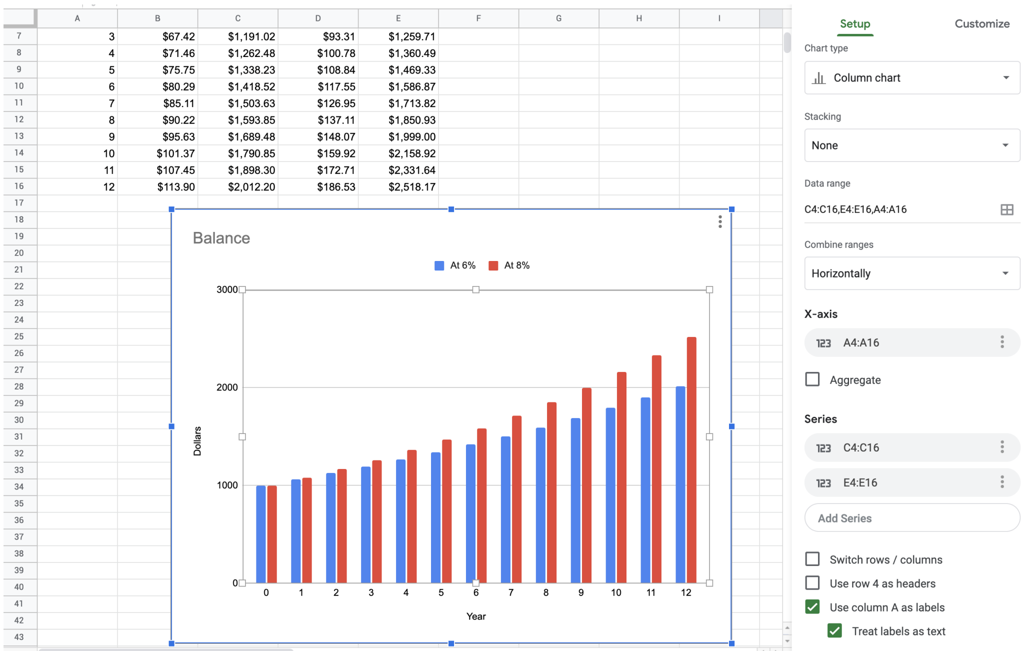


Figure 1.4: A bar graph made in google sheets showing interest.

The year column should be used as the x-axis. There are two series of data that come from C4:C16 and E4:E16. Tidy up the titles and legend as much as you like. Looking at the graph, you can see the balances start the same, but balance of the account with the larger interest rate quickly pulls away from the account with the smaller interest rate.

1.3.2 Line Graph

Here is a line graph:

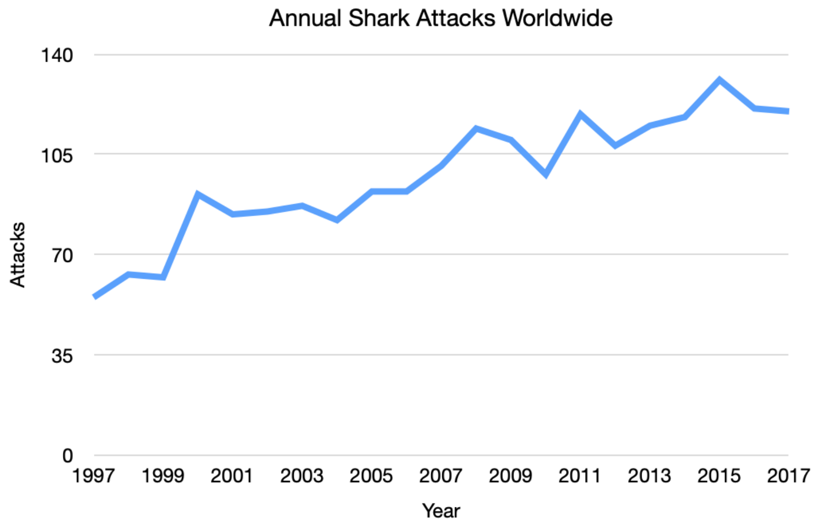


Figure 1.5: A line graph showing shark attacks per year over two decades.

These are often used to show trends over time. Here, for example, you can see that the number of shark attacks has been increasing over time.

You can have more than one line on a graph.

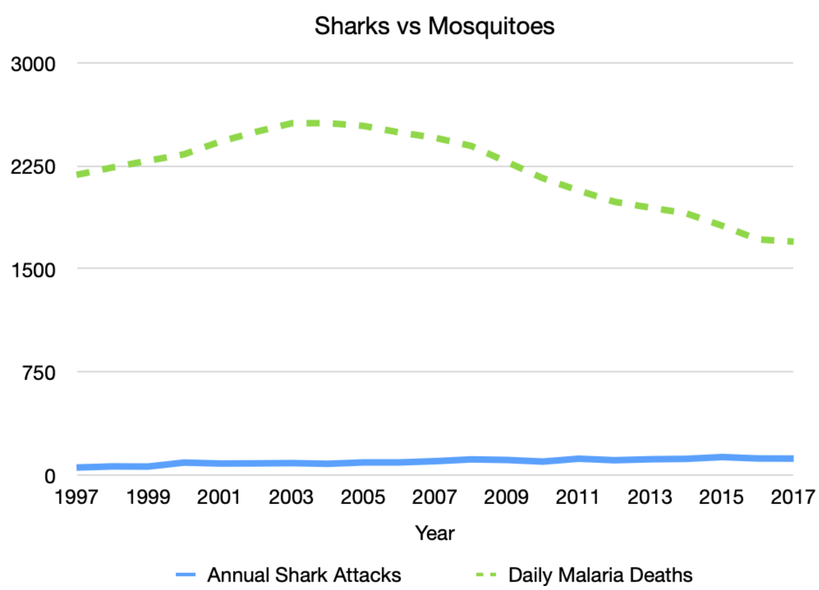


Figure 1.6: A line graph showing shark deaths versus mosquito deaths.

1.3.3 Pie Chart

You use a pie chart when you are looking at the comparative size of numbers. This is best for comparing percentages of a whole that sum to 100%. Here we can see that Nitrogen makes up 78% of the gases in the air.

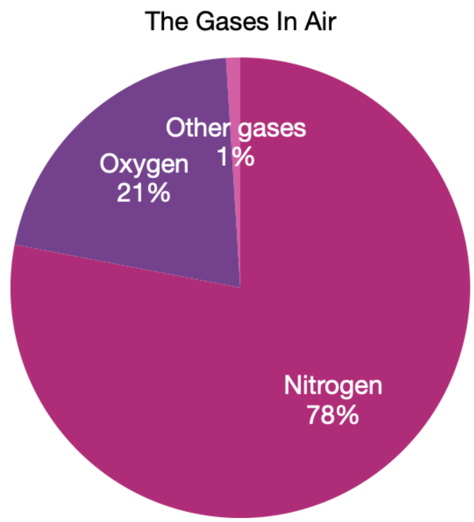


Figure 1.7: A pie chart of the various gases in the air.

When to use a Pie Chart Make a Pie Chart in Python

For *quantitative*, univariate data, we often use dotplots and stemplots to visualize smaller sets of data; and histograms, cumulative frequency plots, and boxplots to visualize larger sets of data. We will explore these types of graphs below.

1.3.4 Dotplot

A *dotplot* is one of the simplest ways to display single, quantitaive variables. Each observation is represented by a dot placed above its value on a number line. If a value appears more than once, the dots stack vertically.

Dotplots are most effective for small to moderate data sets. If the data set is too large, a dotplot can become cluttered and hard to read. In that case, a histogram or a boxplot may be a better choice.

When to Use a Dotplot

- You have 1 variable.

- The variable is quantitative.
- You want to see the distribution and the individual data values.
- The data set is not too large.

How to Make a Dotplot

1. Draw a horizontal number line (x-axis) that covers the data range.
2. Choose a scale that fits the entire range of values.
3. Place one dot above the location of each observation.
4. If multiple observations have the same value, stack the dots vertically.

Here is an example of a dotplot.

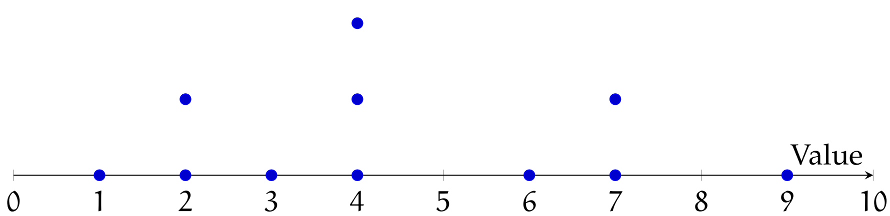


Figure 1.8: A dotplot showing a small sample distribution.

How to Read a Dotplot

Each dot shows the location of one data value. To find the value of a dot, look straight down to the number line.

A dotplot helps you describe the distribution by answering questions like:

- Spread: How spread out are the values (range and clustering)?
- Shape: Do the dots form a roughly symmetric shape, or is the distribution skewed?
- Center: Where is the distribution "balanced" (a typical value)?
- Outliers: Are there values far away from the rest?

Sometimes a dotplot shows a distribution with a long tail to one side. For example, salary data often has many values in the lower-to-middle range and a few extremely high values. This creates a right-skewed distribution.

Outliers can strongly affect numerical summaries, especially the mean.

Connecting Outliers and the Mean (Seesaw Idea) A useful way to think about the mean is as a balancing point. Imagine the dotplot as a seesaw: points far from the main cluster have more "pull." So a single extreme value can shift the mean noticeably, even if most data values stay the same.

Comparative Dotplots

A *comparative dotplot* uses the same numerical scale to compare two (or more) groups. This is useful when you want to compare distributions side-by-side.

When comparing groups, focus on differences in:

- center (typical value),
- spread,
- shape (skew/clusters),
- and outliers.

Make a Dotplot in a Spreadsheet

You can create a dotplot in Excel or Google Sheets using a scatter plot and a helper column.

1. Put your data values in column A (starting in A2). Sort them (recommended).
2. In cell B2, enter:
   `=COUNTIF($A$2:A2, A2)`
   Fill down. This creates stacking heights for repeated x-values.
3. Insert a Scatter chart using column A as x-values and column B as y-values.
4. Format the chart:
   - markers only (no lines),
   - hide the y-axis labels (y is only for stacking),
   - add titles and adjust the x-axis scale.

1.3.5 Stemplot

A *stemplot* (also called a *stem-and-leaf plot*) is an effective way to summarize **one quantitative variable** when the data set is not too large. Like a dotplot, it helps you see the overall distribution, but a stemplot has an extra advantage: it still shows the exact data values.

To make a stemplot, each observation is split into two parts:

- the stem, which is the left-most part of the number (leading digit(s))
- the leaf, which is the remaining part, usually the final digit

For example, 23 can be written as 2 | 3.

How to Make a Stemplot

1. Choose stems and leaves. Decide where to split each number into a stem and a leaf. There is no single rule for this. Your goal is to choose a reasonable number of stems.
2. Draw a vertical divider. Stems go on the left; leaves go on the right.
3. List all stems in order. Make sure the stems cover the full range of your data.
4. Write leaves for each observation. Place each leaf next to its correct stem.
5. Sort leaves. Put the leaves for each stem in increasing order.
6. Include a key. The key shows how to interpret the stems and leaves.

Choosing the Number of Stems

If you use *too few stems*, you can hide patterns because the data get lumped together. If you use *too many stems*, you can also hide patterns because the plot becomes too spread out.

A common strategy is to start with stems that represent groups of 10 (tens digits as stems), and then adjust if needed.

Sometimes a single stem represents too wide of an interval. In that case, you can split stems by repeating each stem twice:

- the first row holds leaves 0–4
- the second row holds leaves 5–9

For example, if your data run from 10 to 49, you can use stems

$$1, 1, 2, 2, 3, 3, 4, 4$$

where the first 1 represents 10–14 and the second 1 represents 15–19, and so on.

Example

Suppose these are housing costs (in dollars per month):

$$255, 259, 304, 306, 309, 317, 323, 327, 332, 334, 342, 349, 350, 351, 362, 367, 376, 384, 401, 403$$

If we use the hundreds digit as the stem and the last two digits as the leaf, we can split each stem into two rows: (top row for leaves 00–49, bottom row for leaves 50–99)

Stem	Leaves
2	55 59
2	
3	04 06 09 17 23 27 32 34 42 49
3	50 51 62 67 76 84
4	01 03
4	

Key: 3 | 04 = 304 dollars per month

How to Read a Stemplot

A stemplot gives you several kinds of information at once:

- Exact values: Each leaf represents a single observation.
- Frequencies: The number of leaves in a row tells how many observations fall in that interval. For example, if a stem row has 9 leaves, that means 9 observations are in that range.
- Gaps: If a stem has no leaves, that means there are no observations in that interval.
- Shape and spread: You can see whether the values cluster in certain ranges and whether the distribution is skewed.
- Center: You can estimate a typical value by locating where the leaves cluster most.
- Outliers: Unusually high or low values may stand out because they appear far from the rest.

Just like with dotplots, you can imagine the distribution as something that could "balance" on a finger: the place it would balance is a rough sense of the center.

Make a Stem Plot in a Spreadsheet Make a Stem Plot in Python

1.3.6 Histogram

A *histogram* is one of the most popular ways to display single quantitaive variables. It is especially useful for showing patterns in larger data sets, where dotplots or stemplots can become overwhelming. A histogram is like a stemplot turned on its side: instead of listing individual values, we group values into intervals called classes or bins, and draw a bar for each bin.

Histograms are best when you care about the overall shape of a distribution (clusters, skew, gaps, outliers), not the exact value of every data point.

A histogram can be drawn using frequency (counts), relative frequency (proportions), and percent frequency (percentages). These three versions have the same shape if they use the same bins. The difference is only the y-axis scale.

How to Make a Histogram

1. Create groups from continuous data. This process is called binning.
2. Draw the x-axis and y-axis, and scale them to fit the class intervals and the frequencies (or relative frequencies or percentages).
3. Draw one bar for each class. The height of each bar equals the frequency (or relative frequency or percent) for that class.
4. Draw the bars touching with no gaps (because the data values are continuous and the bins connect).

How to Read a Histogram

- Each bar represents a single class (bin). There is one bar per class.
- The bins are placed on the x-axis in increasing order, like a number line.
- The height of a bar tells how many observations fall in that class (for a frequency histogram).
- A class with no bar means no observations fall in that interval (a gap).

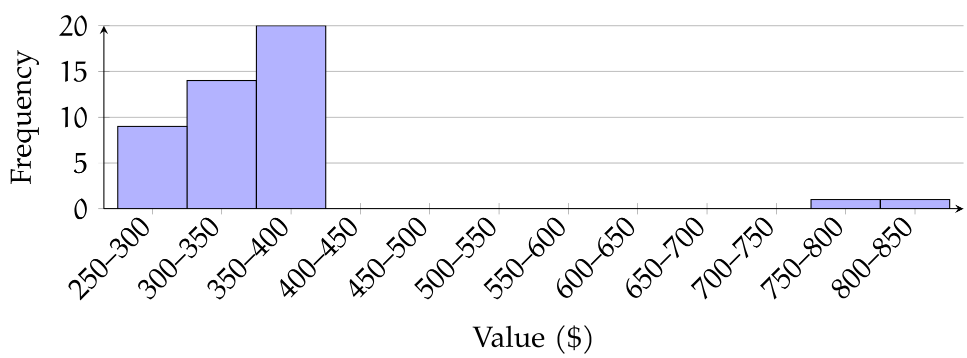


Figure 1.9: A histogram using 50-dollar bins illustrating the example counts (9, 14, 20, ... and 0,1,1 beyond $700).

Example of Interpreting Bars

Suppose a histogram uses the bins \$250–\$300, \$300–\$350, \$350–\$400, and so on.

- If the first bar (250–300) has height 9, then 9 observations are between \$250 and \$300.
- If the second and third bars (300–350 and 350–400) have heights 14 and 20, then $14 + 20 = 34$ observations are between \$300 and \$400.
- If there is no bar for 700–750, then there are 0 observations between \$700 and \$750.
- If the bars beyond 700 have heights 0, 1, and 1, then $0 + 1 + 1 = 2$ observations are at least \$700.

A histogram can help you to describe the distribution by focusing on shape, center, spread, gaps, and outliers.

Making a Histogram in a Spreadsheet

You can make a histogram in Excel or Google Sheets.

1. Put the data in one column.
2. Choose bin widths (for example, bins of width 50: 250–300, 300–350, 350–400, etc.).
3. Insert a histogram chart (or use a column chart after building a frequency table).
4. Check the bin width and the endpoints. Changing the bins can change the appearance of the distribution.

Making a Histogram in Python

Python can generate a histogram quickly and lets you change the bin width easily.

```
import matplotlib.pyplot as plt

# Replace this list with your data
data = [250, 275, 310, 320, 350, 365, 410, 415, 430, 500, 525, 610, 720, 840]

# Choose the number of bins (try changing this!)
plt.hist(data, bins=8)

plt.title("Histogram")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```

Try changing the value of `bins`. Notice that the overall shape can look different depending on how you group the data.

How do we compare dotplots, stemplots, and histograms?

For small data sets, dotplots and stemplots can be better because they show the exact values. In a histogram, you lose that detail because values are grouped into bins. In exchange, histograms make it easier to see the overall pattern in large data sets.

1.3.7 Cumulative Frequency Plot

A *cumulative frequency plot* (also called a *cumulative frequency chart*) shows the running total of how many data values are at or below each value (or at or below each class boundary). Instead of showing how many observations fall in each bin like a histogram, a cumulative frequency plot shows how the totals add up as you move from left to right.

Cumulative frequency plots are useful when you want to answer questions like:

- How many observations are less than or equal to a certain value?
- How many observations are greater than or equal to a certain value?
- What percent of the data fall below a certain cutoff?

How to Make a Cumulative Frequency Plot

1. Draw the x-axis and y-axis.
2. Set up bins for the data and identify the upper class boundaries (the right endpoints).
3. Compute the cumulative frequency at each upper class boundary.
   (This means: add the frequency of that class to the total from all previous bins.)
4. Plot a point at each upper class boundary with height equal to the cumulative frequency.
5. Connect the points with straight line segments.

Interpreting the shape of your plot

The steepness of the curve tells you where the data values are concentrated.

- If the curve rises quickly over a certain x-interval, many observations fall in that range.
- If the curve is almost flat, few observations fall in that range.

A cumulative frequency plot can also hint at skewness:

- **Right-skewed:** the curve increases quickly at smaller values, then levels off later.
- **Left-skewed:** the curve increases slowly at first, then rises sharply near the end.
- **Symmetric:** the curve often looks S-shaped.

Example

Suppose $n = 80$ observations were grouped into bins, and the cumulative frequency at $x = 350$ is about 23. Then:

- About 23 observations are ≤ 350.
- About $80 - 23 = 57$ observations are ≥ 350.

The plot below shows a sample cumulative frequency curve. (These coordinates are an example format; you can replace them with your own cumulative totals.)

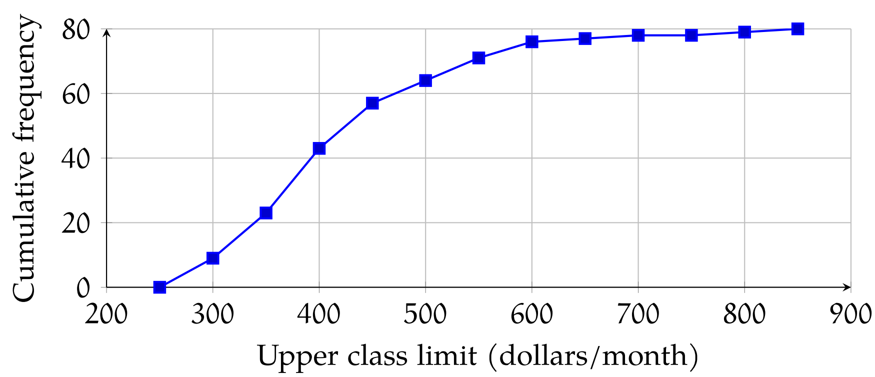


Figure 1.10: A cumulative frequency plot.

Making a Cumulative Frequency Plot in Python

This example shows the basic idea: build bins, count how many values fall in each bin, then take a cumulative sum.

```python
import numpy as np
import matplotlib.pyplot as plt

# Replace with your data
data = np.array([250, 275, 310, 320, 350, 365, 410, 415, 430, 500, 525, 610, 720, 840])

# Choose bin edges (upper class boundaries are the right endpoints)
```

```
bins = np.arange(250, 851, 50)  # 250, 300, ..., 850

# Frequency in each bin
counts, edges = np.histogram(data, bins=bins)

# Cumulative frequency at each upper boundary
cumulative_counts = np.cumsum(counts)
upper_bounds = edges[1:]  # right endpoints

plt.plot(upper_bounds, cumulative_counts, marker='o')
plt.title("Cumulative Frequency Plot")
plt.xlabel("Upper class boundary ($)")
plt.ylabel("Cumulative frequency")
plt.grid(True)
plt.show()
```

If you change the `bin` width, the curve may look different, but it will always increase from left to right and end at n.

Boxplot

1.4 Bivariate Data

1.4.1 Scatter Plot

Sometimes, you have a large number of data points with two values, and you are looking for a relationship between them. For example, maybe you write down the average temperature and the total sales for your lemonade stand on the 15th of every month:

Date	Avg. Temp.	Total Sales
15 January 2022	2.6º C	$183.85
15 February 2022	-4.2º C	$173.56
15 March 2022	13.3º C	$195.22
15 April 2022	26.2º C	$207.61
15 May 2022	27.5º C	$210.88
15 June 2022	31.3º C	$214.18
15 July 2022	33.5º C	$215.23
15 Aug 2022	41.7º C	$224.07
15 September 2022	20.7º C	$198.94
15 October 2022	17.2º C	$196.10
15 November 2022	1.7º C	$185.10
15 December 2022	0.2º C	$188.70

You may wonder, ”Do I sell more lemonade on hotter days?”

To figure this out, you might create a scatter plot. For each day, you put a mark that represents that temperature and the sales that day:

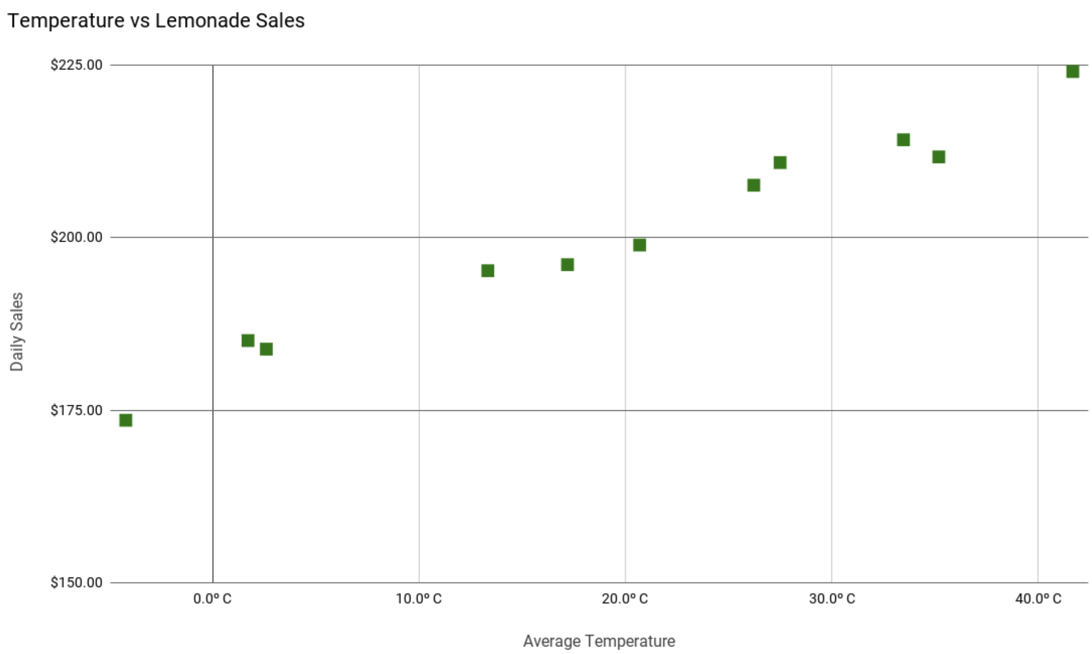


Figure 1.11: A scatterplot of temperature versus daily sales.

From this scatter plot, you can easily see that you do sell more lemonade as the temperature goes up. Drawing a *best-fit* line along the points will give you a *correlation coefficient*. A positive correlation coefficient will give you a positively proportional relationship, while a negative coefficient will give you a inversely proportional relationship.

Making a Scatter Plot in Python

```python
import matplotlib.pyplot as plt

# Avg. Temp (°C) and Total Sales ($)
temps = [2.6, -4.2, 13.3, 26.2, 27.5, 31.3, 33.5, 41.7, 20.7, 17.2, 1.7, 0.2]
sales = [183.85, 173.56, 195.22, 207.61, 210.88, 214.18, 215.23, 224.07, 198.94,
    196.10, 185.10, 188.70]

plt.scatter(temps, sales)
plt.title("Temperature vs. Lemonade Sales")
plt.xlabel("Average Temperature (°C)")
plt.ylabel("Total Sales ($)")
plt.grid(True)
plt.show()
```

Chapter 2

Numerical Methods of Describing Data

In the Introduction to Data Visualization chapter, you learned to read a distribution from a graph. A histogram or boxplot can show you immediately whether data is skewed, where the center sits, and how spread out the values are. But a picture has limits. You cannot subtract two histograms. You cannot use a boxplot to predict a salary or compare a test score to a national average. For that, you need numbers.

This chapter develops the numerical tools that complement the graphical ones you already know. By the end, you will be able to describe any distribution precisely, locate any individual value within it, and avoid the most common errors people make when summarizing data.

2.1 Why Numerical Summaries Matter

Look back at the scatter plot from the previous chapter. To draw the line of best fit through that data, two quantities appeared in the formula: S_x and S_y, the standard deviations of the two variables. The slope of the regression line depends directly on them. Before you can apply any of the methods in the next chapter, you need to know exactly what those quantities are and how to compute them. That is what this chapter is for.

There are three families of numerical measures, each answering a different question about a data set:

- *Measures of central tendency* ask: what is a typical value?
- *Measures of variation* ask: how spread out are the values?
- *Measures of position* ask: where does a particular value sit relative to the rest?

Each family has more than one tool, and choosing the right one depends on the shape of your data. That connection between shape and the choice of numerical summary is one of the central ideas of this chapter.

2.1.1 Summation Notation

Nearly every numerical measure involves adding things up. Before computing anything, we need a compact way to write sums.

Suppose you record the daily sales at a shop over n days, calling them $X_1, X_2, \ldots, X_n$. Writing out every term quickly becomes unwieldy. The Greek capital letter Σ (read as "sigma ") gives us a cleaner way:

$$\sum_{i=1}^{n} X_i = X_1 + X_2 + \cdots + X_n$$

The expression below Σ tells you where to start; the expression above tells you where to stop. When the range is clear from context, we often write $\sum X_i$ or simply $\sum X$.

Consider a student who scores the following on ten quizzes:

$$8,\ 9,\ 0,\ 10,\ 10,\ 8,\ 7,\ 9,\ 10,\ 5$$

Their total score is

$$\sum_{i=1}^{10} X_i = 8 + 9 + 0 + 10 + 10 + 8 + 7 + 9 + 10 + 5 = 76$$

This notation will appear in every formula in this chapter. When you see $\sum X_i$, read it as "add up all the values of X. "

Exercise 1 Summation Practice

A student records the daily high temperatures (in degrees Celsius) over one week: 22, 19, 25, 21, 18, 23, 20.

1. Write the sum of all seven values using summation notation and compute it.

2. Compute $\sum_{i=1}^{7}(X_i - 20)$. What do you notice about how this compares to your answer in (a)?

Working Space

Answer on Page 97

2.2 Measures of Center

You already know how to read the center of a distribution from a graph: it is roughly where the data balances. Now we want to pin that idea down to a single number. There are two main tools for this, and they do not always agree.

2.2.1 The Mean

The *mean* is the arithmetic average of a data set. You can think of it as the balancing point of the distribution. It is the point at which the data would balance if every value were a weight on a number line. This is the same seesaw idea from the graphical chapter, now expressed as a formula.

When the data represent an entire *population* of N values, the mean is denoted by the Greek letter μ (read “mu ”):

$$\mu = \frac{\sum_{i=1}^{N} X_i}{N}$$

When the data are a *sample* of n values drawn from a larger population, the mean is denoted $\bar{X}$ (read "X-bar "):

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

The formula is identical in both cases. Only the notation changes to signal whether you are describing a full population or a sample.

Consider the student from the previous section with ten quiz scores:

$$8,\ 9,\ 0,\ 10,\ 10,\ 8,\ 7,\ 9,\ 10,\ 5$$

$$\bar{X} = \frac{8+9+0+10+10+8+7+9+10+5}{10} = \frac{76}{10} = 7.6$$

Notice what that zero does. One failed quiz pulled the mean down from what would otherwise be a strong set of scores. This is the key property of the mean: it is not *resistant*. Every value in the data set contributes to it, so extreme Values which could be unusually high or low scores, can pull it noticeably away from the center of the bulk of the data.

2.2.2 The Median

The *median* takes a different approach. Rather than averaging all the values, it finds the middle one. Half the data falls at or below the median; half falls at or above it.

To find the median M of a data set with n measurements:

1. Arrange all measurements in increasing order.
2. Compute $l = \frac{n+1}{2}$.
3. The median M is the value of the lth measurement, counted from the smallest.

If n is odd, l is a whole number and the median is a value that actually appears in the data. If n is even, l falls halfway between two observations and the median is their average, it may not correspond to any value in the data.

Return to the car dealership from the summation exercise. Over 14 days, the number of cars sold was:

$$14,\ 9,\ 23,\ 7,\ 11,\ 23,\ 17,\ 11,\ 3,\ 24,\ 21,\ 2,\ 20,\ 20$$

Arranged in order:

$$2,\ 3,\ 7,\ 9,\ 11,\ 11,\ 14,\ 17,\ 20,\ 20,\ 21,\ 23,\ 23,\ 24$$

With $n = 14$:

$$l = \frac{14+1}{2} = 7.5$$

The median is the average of the 7th and 8th observations. Counting from the smallest, the 7th is 14 and the 8th is 17:

$$M = \frac{14+17}{2} = 15.5 \text{ cars}$$

Now notice what happens if the dealership's best day (24 cars) were replaced by an exceptional day of 80 cars. The mean would shift upward. The median would not move at all, as the 7th and 8th values in the sorted list are unchanged. This is what it means for the median to be resistant! That is, it does not react to extreme values the the way the mean does.

Exercise 2 Computing Mean and Median

A small bookshop records the number of books sold each day over eight days:

12, 15, 9, 11, 14, 8, 13, 52

1. Compute the mean.
2. Compute the median.
3. The value 52 is an unusually high day. It could be a sale event. Which measure is more affected by it? Explain why.
4. Remove the value 52 and recompute both the mean and the median using the remaining seven values. How much did each change?

Working Space

Answer on Page 97

2.2.3 Choosing Between Them

The mean and median each have their place, and choosing between them is not arbitrary. The right choice depends on the shape of the distribution, which you can now read directly from a histogram or boxplot.

For a *symmetric distribution*, the mean and median are close to each other. Either works, but the mean is generally preferred because it uses every observation and forms the foundation of most inferential statistics you will encounter later.

For a *skewed distribution* or any data set with outliers, the mean is pulled toward the tail while the median stays near the bulk of the data. In these cases, the median gives a more honest picture of a typical value.

You saw this in *Introduction to Data Visualizations*. Most people earn within a moderate range, but a small number earn vastly more. The mean income for a country is often much higher than what a typical person earns, precisely because it is dragged upward by those extreme high values. The median income is usually the figure reported when the goal is to describe a typical household.

The relationship between mean, median, and skew follows a consistent pattern:

- Symmetric: mean $\approx$ median
- Right-skewed: mean $>$ median
- Left-skewed: mean $<$ median

Exercise 3 Choosing a Measure of Center

For each situation, state whether you would report the mean or the median as the measure of center, and explain your reasoning.

1. The distribution of exam scores in a class of 30 students is roughly symmetric with no outliers.
2. The distribution of monthly salaries at a company is strongly right-skewed. The CEO earns considerably more than all other employees.
3. A histogram of the ages at which people in a city receive their first driving licence shows a large cluster between 16 and 19, and a long tail extending into the 30s and 40s.
4. After removing the CEO's salary from the company data in (b), the remaining distribution is roughly symmetric. Now which measure would you report, and does your answer change?

Working Space

Answer on Page 97

2.3 Measures of Spread

Two data sets can share exactly the same mean and still tell completely different stories. What distinguishes them is spread, which is, how far the values are scattered around the center. This section develops the numerical tools for measuring that scatter precisely.

2.3.1 Range and IQR

You already encountered these in the graphical chapter when building boxplots, so this is a brief review before introducing the more powerful measures.

The *range* is the simplest measure of spread:

$$R = \text{largest value} - \text{smallest value}$$

For the car dealership data, $R = 24 - 2 = 22$ cars. Range is easy to compute but unreliable, because it depends entirely on the two most extreme values and ignores everything in between. One unusual observation can make the range misleading.

The *interquartile range* (IQR) improves on this by focusing on the middle 50% of the data:

$$\text{IQR} = Q_3 - Q_1$$

For the dealership data, $Q_1 = 9$, $Q_3 = 21$, so $\text{IQR} = 12$ cars. Because IQR ignores the tails entirely, it is a resistant measure, so outliers do not affect it. When the median is the appropriate measure of center, IQR is the matching measure of spread.

2.3.2 Variance and Standard Deviation

Range and IQR both have a blind spot: they ignore most of the data. The *standard deviation* fixes this by taking every value into account. It measures how far, on average, each observation sits from the mean.

Before we can compute the standard deviation, we need the *variance*. The variance is the average of the squared distances from the mean. Squaring serves two purposes: it removes the sign so that values above and below the mean do not cancel each other out, and it penalizes large deviations more heavily than small ones.

The *population variance* is denoted σ^2:

$$\sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$$

The *sample variance* is denoted s^2:

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$$

We divide by $n - 1$ rather than n when working with a sample. Dividing by n would systematically underestimate the true population spread because a sample tends to cluster closer to its own mean than the full population does. Subtracting one corrects for that.

The standard deviation is simply the square root of the variance, which brings the units back to the same scale as the original data:

$$\sigma = \sqrt{\sigma^2} \qquad\qquad s = \sqrt{s^2}$$

A population standard deviation is denoted σ ("sigma "); a sample standard deviation is denoted s.

Computing Standard Deviation by Hand

Let us work through the quiz score data step by step. The ten scores are $8, 9, 0, 10, 10, 8, 7, 9, 10, 5$ with $\bar{X} = 7.6$.

Step 1. Compute each deviation from the mean, $x_i - \bar{x}$:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
8	0.4	0.16
9	1.4	1.96
0	−7.6	57.76
10	2.4	5.76
10	2.4	5.76
8	0.4	0.16
7	−0.6	0.36
9	1.4	1.96
10	2.4	5.76
5	−2.6	6.76
		86.40

Step 2. Divide the sum of squared deviations by $n - 1 = 9$ to get the sample variance:

$$s^2 = \frac{86.40}{9} = 9.60$$

Step 3. Take the square root:

$$s = \sqrt{9.60} \approx 3.10$$

Notice that the zero score contributes a squared deviation of 57.76, which is larger than the contributions of all nine other scores combined. One extreme value had an outsized effect on the standard deviation. This is what it means for standard deviation to be non-resistant.

Exercise 4 Computing Variance and Standard Deviation

A small car dealership records the number of cars sold each day over 14 days:

14, 9, 23, 7, 11, 23, 17, 11, 3, 24, 21, 2, 20, 20

The mean is $\bar{X} \approx 14.64$ cars.

1. The sum of squared deviations $\sum(x_i - \bar{x})^2$ for this data set is approximately 800.38. Use this to compute the sample variance s^2.
2. Compute the sample standard deviation s.
3. Standard deviation is measured in the same units as the original data. What are the units of s here? What are the units of s^2?
4. A second dealership has a mean of 14.64 cars per day and $s = 2.1$ cars. Without any other information, which dealership has more consistent daily sales? Explain.

Working Space

Answer on Page 98

2.3.3 Interpreting Spread

Standard deviation is a unit of distance. Once you have computed s for a data set, you can ask of any individual observation: how many standard deviations away from the mean is this value? That question will become the foundation of the next section on z-scores.

For now, focus on two properties that are always true.

First, a larger standard deviation means a wider distribution. Compare two students' quiz scores:

$$\begin{array}{lll} \text{Student A:} & 8,\ 9,\ 4,\ 8,\ 6,\ 8,\ 7,\ 9,\ 10,\ 5 & s_A \approx 1.90 \\ \text{Student B:} & 7,\ 8,\ 6,\ 7,\ 6,\ 6,\ 7,\ 5,\ 5,\ 7 & s_B \approx 0.97 \end{array}$$

Student A's scores vary more from quiz to quiz. Student B's are tightly packed. The standard deviations reflect exactly what you see in the data: $s_A > s_B$.

Second, standard deviation is not resistant. Because it is computed from the mean, and the mean is pulled by extreme values, any outlier also inflates the standard deviation. Recall the quiz example: the single zero score contributed a squared deviation of 57.76 out of a total of 86.40. One observation accounted for two-thirds of the total spread.

This is why the choice of spread measure mirrors the choice of center measure:

- Symmetric data, no outliers: use mean and standard deviation
- Skewed data or data with outliers: use median and IQR

Exercise 5 Choosing a Measure of Spread

For each situation, state which measure of spread (standard deviation or IQR) is more appropriate, and explain why.

1. The distribution of heights of adult men in a country is roughly symmetric and bell-shaped with no outliers.
2. The distribution of house prices in a city is strongly right-skewed, with a small number of luxury properties selling for many times more than the typical home.
3. A researcher removes all outliers from a right-skewed data set. The remaining distribution is approximately symmetric. Which measure of spread should she now report?

Working Space

Answer on Page 98

2.4 Connecting Shape to Numerical Measures

You already know how to read shape from a histogram or boxplot. This section translates what you see graphically into numerical consequences: specifically, what skew and outliers do to the measures you just computed.

2.4.1 Symmetric Data: Mean Approximately Equals Median

When a distribution is roughly symmetric, the mean and median land close to each other. Neither tail is pulling the mean away from the center of the data. This is the situation where mean and standard deviation are both appropriate, and where most inferential procedures work as intended.

2.4.2 Skewed Data: Mean Pulled Toward the Tail

When a distribution is skewed, the mean chases the tail while the median stays near the bulk of the data. The direction of the skew tells you exactly which way the mean is pulled:

- Right-skewed: mean > median. The right tail pulls the mean upward.
- Left-skewed: mean < median. The left tail pulls the mean downward.

This is not just a rule to memorize! It also follows directly from what you know about the mean. Because the mean uses every value in its calculation, a cluster of extreme values in the right tail will drag it rightward. The median, which depends only on rank order, does not move.

Consider a small right-skewed data set:

$$2,\ 3,\ 4,\ 4,\ 5,\ 5,\ 5,\ 6,\ 6,\ 7,\ 15,\ 20$$

The mean is approximately 6.83 and the median is 5.00. The two extreme values at 15 and 20 pull the mean well above where most of the data sits. On a histogram, most bars would be clustered at the left with a long right tail. This shape corresponds to mean > median.

2.4.3 Effects of Outliers

You have heard this term used all over our Statistics chapters! An *outlier* is a value that sits far from the rest of the data. You saw the numerical definition in the graphical chapter: any value more than $1.5 \times \text{IQR}$ below Q_1 or above Q_3 is flagged as an outlier.

What outliers do to each measure follows directly from whether that measure is resistant:

- Mean: pulled toward the outlier. One extreme value can move the mean substantially.
- Median: unaffected. Moving the most extreme value further into the tail does not change the middle rank.
- Standard deviation: inflated. Because it is computed from the mean, an outlier that moves the mean also increases the squared deviations.
- IQR: unaffected. It depends only on Q_1 and Q_3, which are determined by the middle of the data.

This is why mean and standard deviation always travel together, and median and IQR always travel together. They are matched pairs, both resistant or both non-resistant.

2.4.4 Changing Units on Summary Measures

One practical question comes up often: if every value in a data set is shifted or scaled, what happens to the summary measures? Suppose a teacher adds 20 points to every student's test score, or a government increases all property taxes by 2%. You do not need to recompute everything from scratch.

The rules depend on whether you are adding a constant or multiplying by one.

Adding a constant a to every value. Shifting every observation by the same amount shifts the measures of center and position by the same amount. Measures of spread are unaffected, because the distances between values do not change.

Multiplying every value by a constant b. Scaling every observation multiplies all measures, such as center, spread, and position, by the same factor. The absolute value $|b|$ is used for spread measures, since spread is always non-negative.

These rules are summarized in the table below.

Measure	$Y_i = X_i + a$	$Y_i = b \cdot X_i$
Mean	$\bar{Y} = \bar{X} + a$	$\bar{Y} = b\bar{X}$
Median	$\text{Median}(Y) = \text{Median}(X) + a$	$\text{Median}(Y) = b \cdot \text{Median}(X)$
Range	Unaffected	$\text{Range}(Y) = \lvert b \rvert\, \text{Range}(X)$
Std dev	Unaffected	$s_Y = \lvert b \rvert \cdot s_X$
Q_1, Q_3	$Q_1 + a$, $Q_3 + a$	$b \cdot Q_1$, $b \cdot Q_3$
IQR	Unaffected	$\text{IQR}(Y) = \lvert b \rvert \cdot \text{IQR}(X)$

The key pattern: adding a constant moves everything but changes no distances; multiplying stretches or compresses everything including distances.

Example. A class takes a five-question test worth 20 points each. The summary statistics for their scores are:

Mean	62
Median	60
Range	45
Standard deviation	8
Q_1	48
Q_3	71
IQR	23

After grading, the teacher discovers a typographical error in question 2 that made it unanswerable. She adds 20 points to every student's score. The new summary statistics are:

Measure	Original	After adding 20
Mean	62	$62 + 20 = 82$
Median	60	$60 + 20 = 80$
Range	45	45 (unaffected)
Std dev	8	8 (unaffected)
Q_1	48	$48 + 20 = 68$
Q_3	71	$71 + 20 = 91$
IQR	23	23 (unaffected)

Every student's score shifted by the same amount, so the spread of the distribution is unchanged. The center and quartiles each moved up by exactly 20.

Exercise 6 Changing Units

County residents vote to increase property taxes by 2% to fund local schools. The current summary statistics for property tax (in dollars per year) are:

Mean	12,000
Median	8,000
Range	30,000
Standard deviation	5,000
Q_1	5,000
Q_3	14,000
IQR	9,000

Each resident's new tax will be $1.02\times$ their current tax. Compute the new value of each summary measure.

Working Space

Answer on Page 98

2.5 Z-scores

The previous section showed that adding a constant to every value shifts the mean but leaves the standard deviation unchanged. That means the distances between values, relative to the center, stay exactly the same. There is a natural way to express those distances as a single number: the *z-score*, also called a *standardized score*.

The z-score of a measurement tells you how far that value sits from the mean, expressed in units of standard deviations:

$$z = \frac{x - \bar{x}}{s}$$

For population data, replace $\bar{x}$ with μ and s with σ. The interpretation is the same either way.

A positive z-score means the value is above the mean. A negative z-score means it is below. A z-score of zero means the value equals the mean exactly. The magnitude tells you how many standard deviations separate the value from the center.

Example. A class takes a test with mean 74 and standard deviation 6. Two students receive the following scores:

$$\begin{array}{lll} \text{Student A} & 88 & z = \dfrac{88 - 74}{6} = \dfrac{14}{6} \approx 2.33 \\ \text{Student B} & 70 & z = \dfrac{70 - 74}{6} = \dfrac{-4}{6} \approx -0.67 \end{array}$$

Student A scored about 2.33 standard deviations above the class average. Student B scored about 0.67 standard deviations below it. Student B is actually closer to the mean than Student A: $|-0.67| < |2.33|$.

Comparing Across Different Distributions

Z-scores become especially powerful when comparing measurements from distributions with different means and standard deviations. Without standardization, comparing values measured on different scales is meaningless. With z-scores, it becomes straightforward.

Suppose two cities report their daily high temperatures:

City	Mean daily high	Standard deviation
North Bend	80°F	12°F
South Bend	84°F	4°F

Both cities report a high of 95°F on the same day. Which city had the more unusually warm day?

The raw difference from the mean is larger for North Bend ($95-80=15$) than South Bend ($95-84=11$), but that ignores how variable the temperatures are in each city. Compute the z-scores:

$$\text{North Bend: } z=\frac{95-80}{12}=1.25 \qquad \text{South Bend: } z=\frac{95-84}{4}=2.75$$

South Bend's 95°F sits 2.75 standard deviations above its Mean which is far more unusual than North Bend's 1.25. The day was more remarkably warm in South Bend, even though it was not further above South Bend's mean in raw degrees.

Exercise 7 Computing and Interpreting Z-scores

The car dealership from earlier in this chapter had mean daily sales of $\bar{X} \approx 14.64$ cars and standard deviation $s \approx 7.56$ cars.

1. Compute the z-score for the best sales day (24 cars). Interpret it in context.
2. Compute the z-score for the worst sales day (2 cars). Interpret it in context.
3. A second dealership reports a best day of 31 cars, with mean 20 and standard deviation 4. Which dealership had the more unusually good best day? Show your work.
4. The owner of the first dealership adds a fixed weekend bonus, increasing every daily total by 5 cars. Without recomputing, what is the z-score for the original best day of 24 cars under the new totals? Explain.

Working Space

Answer on Page 99

Key Points:

- A z-score expresses a value as a number of standard deviations above or below the mean
- Positive z-scores are above the mean; negative z-scores are below it
- Z-scores allow meaningful comparison across data sets measured on different scales
- Adding a constant to every value does not change any z-score; z-scores are invariant under shifting

2.6 Percentiles and Quartiles

The mean, median, and standard deviation describe the whole distribution. Sometimes the question is narrower: where does one particular value sit relative to everyone else? A student who scores 1400 on the SAT wants to know not just that the median was 1040, but what percentage of students she scored above. That is what percentiles and quartiles answer.

Percentiles

The k*th percentile* P_k is the value such that $k\%$ of observations fall at or below it and $(100-k)\%$ fall at or above it. For example, P_{95} is the 95th percentile: a value at P_{95} is higher than 95% of the data and lower than the remaining 5%.

Three percentiles you already know appear here under new names: $P_{25} = Q_1$, $P_{50} = Q_2 = M$ (the median), and $P_{75} = Q_3$.

To find P_k for a data set of n measurements:

1. Arrange all measurements in increasing order.
2. Compute $l = \dfrac{(n+1)k}{100}$.
3. P_k is the value of the measurement in position l, counted from the smallest. If l is not a whole number, interpolate between the two surrounding observations.

Return to the dealership data ($n = 14$, sorted):

$$2,\ 3,\ 7,\ 9,\ 11,\ 11,\ 14,\ 17,\ 20,\ 20,\ 21,\ 23,\ 23,\ 24$$

To find P_{90}:

$$l = \frac{(14+1)\times 90}{100} = 13.5$$

The 90th percentile falls halfway between the 13th and 14th observations. The 13th is 23 and the 14th is 24:

$$P_{90} = \frac{23+24}{2} = 23.5 \text{ cars}$$

On 90% of days, the dealership sold 23.5 or fewer cars.

To find P_{10}:

$$l = \frac{(14+1)\times 10}{100} = 1.5$$

Halfway between the 1st observation (2) and the 2nd (3):

$$P_{10} = \frac{2+3}{2} = 2.5 \text{ cars}$$

On 10% of days, the dealership sold 2.5 or fewer cars.

Quartiles and the Five-Number Summary

Quartiles divide a sorted data set into four equal parts using the 25th, 50th, and 75th percentiles. You first encountered these when building boxplots in the *Introduction to Data Visualization* chapter. Here is the formal connection:

- $Q_1 = P_{25}$: 25% of values fall at or below Q_1
- $Q_2 = P_{50} = M$: the median
- $Q_3 = P_{75}$: 75% of values fall at or below Q_3

In practice, quartiles are computed by splitting the sorted data into a lower half and an upper half and taking the median of each. For even n, split the data at the midpoint with no overlap. For odd n, exclude the median from both halves.

For the dealership data, split at the midpoint ($n = 14$, so seven values on each side):

$$\text{Lower half: } 2,\ 3,\ 7,\ \mathbf{9},\ 11,\ 11,\ 14 \Rightarrow Q_1 = 9$$
$$\text{Upper half: } 17,\ 20,\ 20,\ \mathbf{21},\ 23,\ 23,\ 24 \Rightarrow Q_3 = 21$$

You may notice that the median-of-halves method gives $Q_1 = 9$, while the percentile formula gives $P_{25} = 8.5$. Both are correct, as different software and textbooks use slightly different conventions for quartile computation. Small disagreements between methods are normal and expected. On the AP exam, the median-of-halves method is standard.

Together with the minimum and maximum, the quartiles form the *five-number summary*:

the same five values you used to build a boxplot.

Min	Q_1	Median	Q_3	Max
2	9	15.5	21	24

The five-number summary and the boxplot are two representations of exactly the same information: one numerical, one graphical.

Exercise 8 Relative Standing

Use the dealership data:
2, 3, 7, 9, 11, 11, 14, 17, 20, 20, 21, 23, 23, 24 $(n = 14)$.

1. Compute P_{50} using the formula $l = \frac{(n+1)k}{100}$. Does it match the median you computed earlier? What position in the sorted list does l point to?
2. On what percentage of days did the dealership sell at most 21 cars?
3. A sales consultant tells the owner that a day with 17 cars sold falls in the "third quarter " of the distribution. Verify this by identifying which quarter 17 belongs to.
4. The owner wants to flag the bottom 10% of days as underperforming. Using $P_{10} = 2.5$ cars, how many of the 14 days fall in this category? List them.

Working Space

Answer on Page 99

Chapter 3

Introduction to Graphs

3.1 Introduction

Some data is easier to work with if we imagine it as a set of connections. When we say *graph* in this chapter, we are referring to set of data defined by a collection of *nodes* (you may also hear vertex) and connected by *edges*. For example, on some social networks, each user can follow any number of other users. We can think of each user as a node, and the edge points from the user who follows to the user they follow:

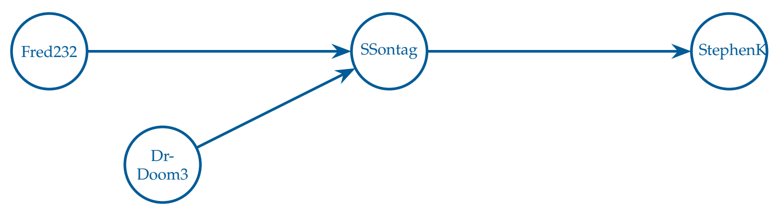


Figure 3.1: A graph showing four users and three "following" relationships.

Following is a *directed* relationship: Fred232 follows SSontag, but SSontag doesn't follow Fred232. So we would say that this is a *directed graph* with four nodes and three edges.

There are also undirected graphs. For example, you can imagine a graph that represents big data lines between cities. All the big data lines allow communications in both directions, like two-way roads:

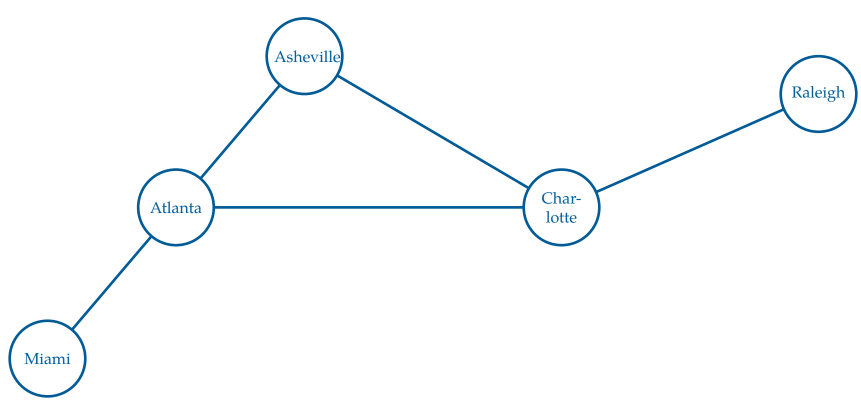


Figure 3.2: A set of cities, represented by nodes, on the east coast of the United States connected by roads, represented by edges.

The arrows are gone; if data can flow from Miami to Raleigh, then data can flow from Raleigh to Miama. A linked connection of edges between two or more nodes is called a *path*. Notice how Atlanta and Charlotte have 3 edges; we would say that Atlanta and Charlotte each have a *degree* of 3.

There is a whole branch of mathematics called *Graph Theory* that studies the properties of graphs. Here are two questions that mathematicians might ask about this graph:

- What is the smallest number of edges that we would need to follow to get from Miami to Raleigh?
- Does the graph have any paths where you could end up where you started?

That second question asks if there is a *cycle* in the graph. This graph has one cycle: Atlanta $\rightarrow$ Asheville $\rightarrow$ Charlotte $\rightarrow$ Atlanta.

Some graphs are *connected*: You can get from one node to any other node by following edges. Is this graph connected?

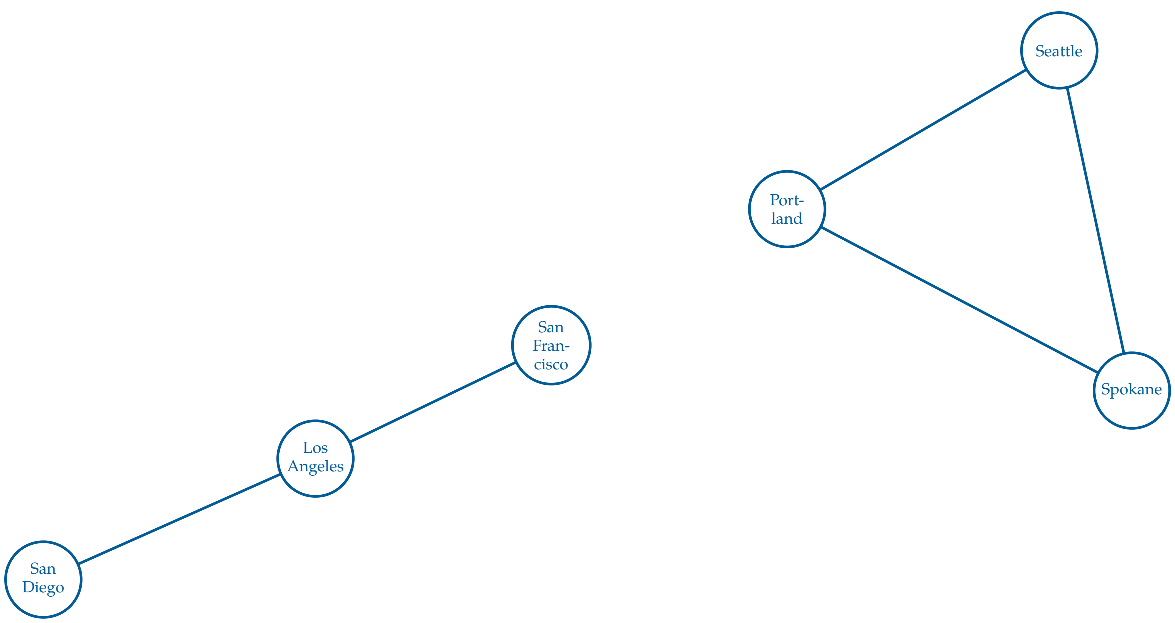


Figure 3.3: A set of cities (nodes) that are not all connected. Each set of *connected* cities forms a *connected component*.

This graph is *not* connected! You can't follow edges from San Diego to Seattle. However, you could identify two *connected components*, *subgraphs* of a graph that remain connected.

- San Diego, Los Angeles, San Francisco
- Portland, Seattle, Spokane

Connectivity is defined as, for all pairs of nodes in that graph, there exists a path between them. A subgraph is much simpler than a connected component, it's only requirement is to contain a subset of at least one node and, optionally, edges to go between nodes. Note that you cannot have a subgraph that contains a single node and an edge, because the edge would not be able to connect to anything else.

In graph data, the nodes and edges often have attributes. For example, a node representing a city might have a name and a population. An edge representing a data line might have a bandwidth (bits per second) and a latency (how many nanoseconds between when you put a bit into the pipe and when it comes out the other end). This is how we define a graph.

Exercise 9 Word Graphs

Working Space

Consider the concept of a graph as a set of nodes and edges. How would you create a graph where the nodes are words and the edges represent a one letter difference between two words?

For example, the word `cat` would be connected to `cot`, `cart`, and `cut`, but not to `dog`.

Define explciitly what the nodes would be in this graph, and what the edges would be. Draw a simple example of this graph with at least 5 nodes and 5 edges.

Answer on Page 100

3.2 Finding Good Paths

For many problems, we are trying to find the best path from one node to another. If all the edges are the same, this usually means finding the path that requires walking the fewest edges.

Sometimes the edges have a cost attribute. For example, you might want to find the cheapest way to ship a container from New York City to Long Beach, California. In this case the nodes are train depots. Each train line between the depots has a cost. What is the cheapest path?

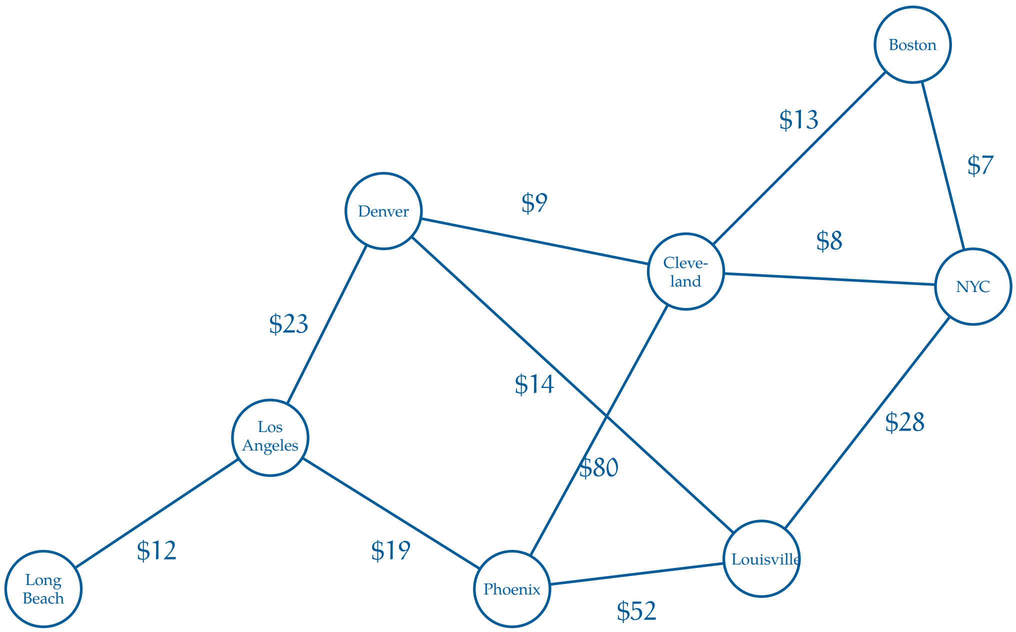


Figure 3.5: A graph of interconnected cities where each edge has a respective cost.

When edges have costs like this, we call the *weighted edges*, which forms a *weighted graph*. In weighted graphs, the *degree* of a node is split into the *in-degree* and *out-degree* of nodes. The in-degree identifies the incoming edges that originate from other nodes, and out-degree is the amount of edges going out of that node.

The graphs that you see here are really small, so finding efficient paths isn't difficult — you could just try all of them! However, in many computer programs, we are working with *millions* of nodes and edges. Efficient graph algorithms are *really* important.

3.3 Graphs in Python

There are many ways to represent graph data. There are database systems that are specifically designed to hold and analyze graph data. Not surprisingly, these are called *Graph Databases*.

You may also hear about *adjacency lists*, which store, for each node, the list of nodes it is connected to. Each direct connection from one node to another is called a *neighbor*. Simply, adjacency lists store lists of neighbors.

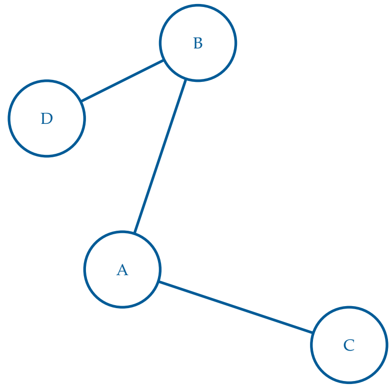


Figure 3.6: A simple (undirected unweighted) graph used to make an adjacency list.

As an adjacency list, this would look like:

- A: [B, C]
- B: [A, D]
- C: [A]
- D: [B]

Exercise 10 Directed Adjacency Lists

Consider the concept of adjacency lists discussed above. How would the storage method change if the edges were to have weights?

Working Space

Answer on Page 100

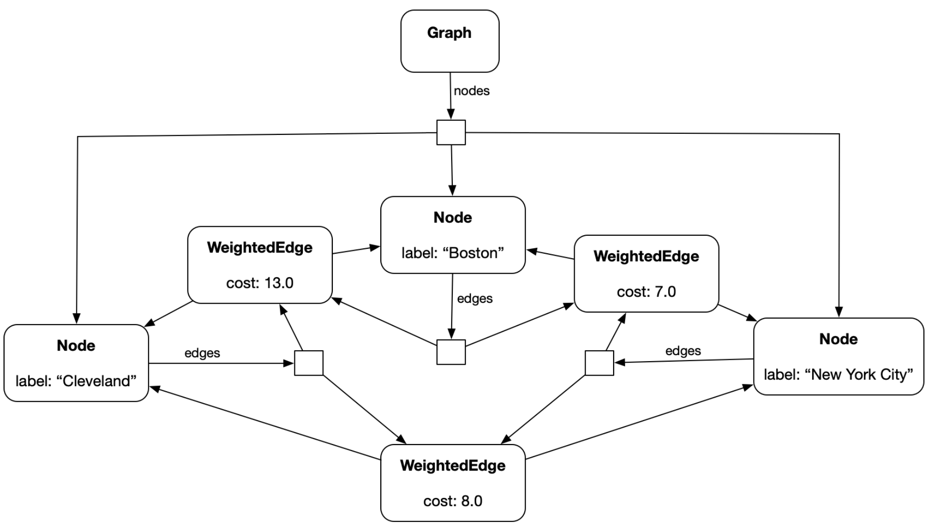
For simplicity, you are going to write Python classes that will let you represent an undirected graph with weighted edges, like the shipping problem above.

(Naturally, things would look a little different if the graph were directed or the edges were unweighted, but this is a good starting place.)

Create a file called `graph.py`. This will hold the code for your `Node` and `WeightedEdge` classes. We will also create a `Graph` class that will just hold onto the list of its nodes.

- A `Node` will have a label string and a list of edges that touch it.
- A `WeightedEdge` will have a cost and two nodes: `node_a` and `node_b`.
- A `Graph` will have a list of nodes.

Here is what the object diagram would look like if you had only three cities:



Put this code into `graph.py`

```
class Node:
  def __init__(self, label):
    self.label = label
    self.edges = []

  def __repr__(self):
    return f"(node:{self.label}, edges:{len(self.edges)})"

class WeightedEdge:
  def __init__(self, cost, node_a, node_b):
    self.cost = cost
    self.node_a = node_a
    node_a.edges.append(self)
    self.node_b = node_b
    node_b.edges.append(self)

```

```python
def other_end(self, node_from):
    if self.node_a == node_from:
        return self.node_b
    if self.node_b == node_from:
        return self.node_a
    raise ValueError("node is not an endpoint of this edge")

class Graph:
  def __init__(self):
    self.nodes = []

  def add_node(self, new_node):
    self.nodes.append(new_node)

  def __repr__(self):
    return f"(Graph:{self.nodes})"
```

This is a graph class that handles undirected weighted graphs. Now, let's create some instances of `Node` and `WeightedEdge` and wire them together. Create another file in the same directory called `cities.py`. Put in this code:

```python
import graph

# Create an empty graph
network = graph.Graph()

# Create city nodes and add to graph
long_beach = graph.Node("Long Beach")
network.add_node(long_beach)
los_angeles = graph.Node("Los Angeles")
network.add_node(los_angeles)
denver = graph.Node("Denver")
network.add_node(denver)
phoenix = graph.Node("Phoenix")
network.add_node(phoenix)
louisville = graph.Node("Louisville")
network.add_node(louisville)
cleveland = graph.Node("Cleveland")
network.add_node(cleveland)
boston = graph.Node("Boston")
network.add_node(boston)
nyc = graph.Node("New York City")
network.add_node(nyc)

# Create edges
graph.WeightedEdge(12, long_beach, los_angeles)
graph.WeightedEdge(23.0, los_angeles, denver)
graph.WeightedEdge(19, los_angeles, phoenix)
graph.WeightedEdge(52, phoenix, louisville)
graph.WeightedEdge(14, denver, louisville)
```

```
graph.WeightedEdge(80, phoenix, cleveland)
graph.WeightedEdge(9, denver, cleveland)
graph.WeightedEdge(8, cleveland, nyc)
graph.WeightedEdge(28, louisville, nyc)
graph.WeightedEdge(7, nyc, boston)
graph.WeightedEdge(13, cleveland, boston)

print(network)
```

Run it:

```
python3 cities.py
```

You should see some rather unexciting output:

```
(Graph:[(node:Long Beach, edges:1), (node:Los Angeles, edges:3), (node:Denver,
↪  edges:3),
(node:Phoenix, edges:3), (node:Louisville, edges:3), (node:Cleveland, edges:4),
(node:Boston, edges:2), (node:New York City, edges:3)])
```

But do not worry; we will make it more exciting in the next chapter!

Chapter 4

The Dot Product

If you have two vectors $\mathbf{u} = [u_1, u_2, \ldots, u_n]$ and $\mathbf{v} = [v_1, v_2, \ldots, v_n]$, we define the *dot product* $\mathbf{u} \cdot \mathbf{v}$ as

$$\mathbf{u} \cdot \mathbf{v} = (u_1 \times v_1) + (u_2 \times v_2) + \cdots + (u_n \times v_n)$$

The output of the dot product is a *scalar* quantity.

For example,

$$[2, 4, -3] \cdot [5, -1, 1] = 2 \times 5 + 4 \times -1 + -3 \times 1 = 3$$

This may not seem like a very powerful idea, but dot products are *incredibly* useful. The enormous GPUs (Graphics Processing Units) that let video games render scenes so quickly? They primarily function by computing huge numbers of dot products at mind-boggling speeds.

Exercise 11 Basic dot products

Compute the dot product of each pair of vectors:

- $[1, 2, 3]$, $[4, 5, -6]$
- $[\pi, 2\pi]$, $[2, -1]$
- $[0, 0, 0, 0]$, $[10, 10, 10, 10]$

Working Space

Answer on Page 101

4.1 Properties of the dot product

Sometimes we need an easy way to say "The vector of appropriate length is filled with zeros." We use the notation $\vec{0}$ to represent this. Then, for any vector $\mathbf{v}$, this is true:

$$\mathbf{v} \cdot \vec{0} = 0$$

The dot product is commutative:

$$\mathbf{v} \cdot \mathbf{u} = \mathbf{u} \cdot \mathbf{v}$$

The dot product of a vector with itself is its magnitude squared:

$$\mathbf{v} \cdot \mathbf{v} = |\mathbf{v}|^2$$

If you have a scalar a, then:

$$(\mathbf{v}) \cdot (a\mathbf{u}) = a(\mathbf{v} \cdot \mathbf{u})$$

So, if **v** and **w** are vectors that go in the same direction,

$$\mathbf{v} \cdot \mathbf{w} = |\mathbf{v}||\mathbf{w}|$$

If **v** and **w** are vectors that go in opposite directions,

$$\mathbf{v} \cdot \mathbf{w} = -|\mathbf{v}||\mathbf{w}|$$

If **v** and **w** are vectors that are perpendicular to each other, their dot product is zero:

$$\mathbf{v} \cdot \mathbf{w} = 0$$

4.2 Cosines and dot products

Furthermore, dot products' interaction with cosine makes them even more useful is what makes them so useful: If you have two vectors **v** and **u**,

$$\mathbf{v} \cdot \mathbf{u} = |\mathbf{v}||\mathbf{u}| \cos \theta$$

where θ is the angle between them.

So, for example, if two vectors $\mathbf{v}$ and $\mathbf{u}$ are perpendicular, the angle between them is $\pi/2$. The cosine of $\pi/2$ is 0. The dot product of any two perpendicular vectors is always 0. In fact, if the dot product of two non-zero vectors is 0, the vectors *must be* perpendicular (see figure 4.1 for an example of perpendicular 2-dimensional vectors).

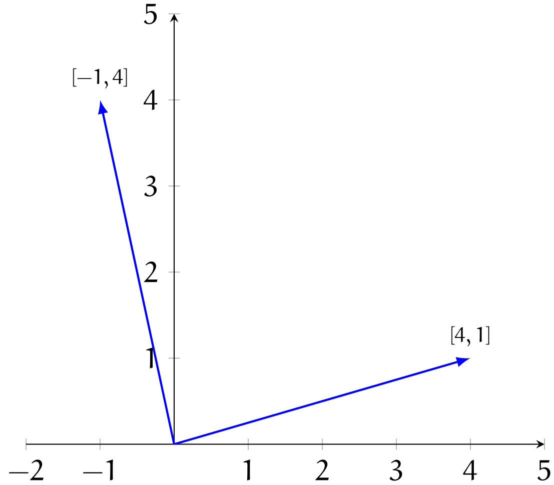


Figure 4.1: The dot product of any two perpendicular vectors is zero.

If you have two non-zero vectors $\mathbf{v}$ and $\mathbf{u}$, you can always compute the angle between them:

$$\theta = \arccos\left(\frac{\mathbf{v} \cdot \mathbf{u}}{|\mathbf{v}||\mathbf{u}|}\right)$$

Arccos is short for arccosine, or $\cos^{-}1$, and it is a function that is the inverse of cosine. Cosine takes an angle and gives back the scaled x-component of the angle. Arccosine takes the x-component of an angle and returns an angle with that x-component. However, there is a limit to what arccos can return. Let's look at cosine and its inverse, arccos (see figures 4.2 and 4.3).

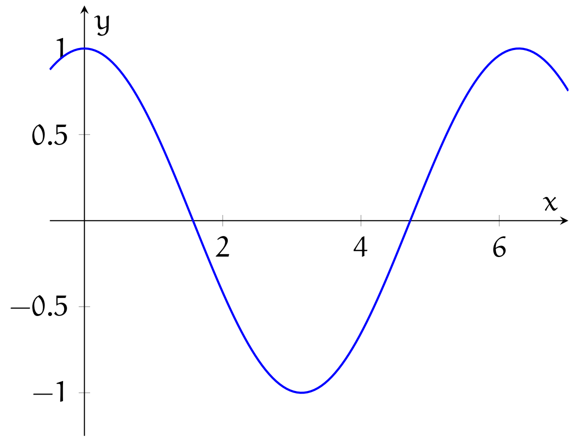


Figure 4.2: Cosine is a function: there is exactly one output for every input.

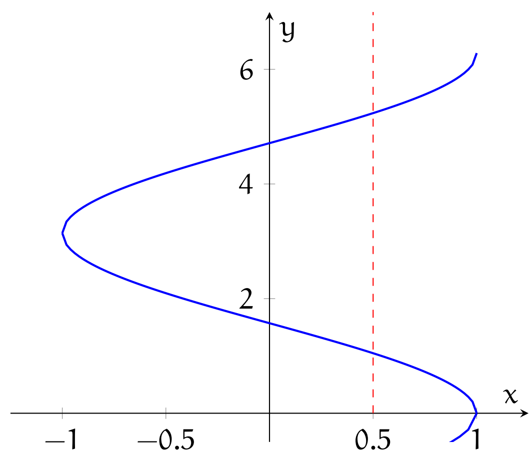


Figure 4.3: Arccos is not a function: there are many angles with the same x-component. Notice that one input value has many output values (see the red dashed line).

When you use a calculator to evaluate arccos, the calculator automatically restricts the results to between 0 and π. Let's look at an example of using the dot product to determine the angle between two vectors:

Example: What is the angle between $\mathbf{u} = \left[\sqrt{3}, 1\right]$ and $\mathbf{v} = [0, -1]$?

Solution: We know that $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}|\,|\mathbf{v}| \cos\theta$. Therefore, we also know that:

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}|\,|\mathbf{v}|}$$

First, let's compute the dot product:

$$\mathbf{u} \cdot \mathbf{v} = \sqrt{3} \cdot 0 + 1 \cdot -1 = -1$$

And therefore:

$$\cos\theta = \frac{-1}{|\mathbf{u}|\,|\mathbf{v}|}$$

Now, let's find the magnitudes of both vectors:

$$|\mathbf{u}| = \sqrt{\left(\sqrt{3}\right)^2 + (1)^2} = 2$$

$$|\mathbf{v}| = \sqrt{(0)^2 + (-1)^2} = 1$$

Substituting for the magnitudes, we find that:

$$\cos\theta = \frac{-1}{2 \cdot 1} = \frac{-1}{2}$$

To solve for θ, we take the arccos of both sides:

$$\arccos(\cos\theta) = \theta = \arccos\frac{-1}{2}$$

What angles have a cosine of $-1/2$? We know that $2\pi/3$, $4\pi/3$, $8\pi/3$, etc., all have a cosine of $-1/2$. Because the range of arccos is restricted to between 0 and π, our result is:

$$\theta = \arccos\frac{-1}{2} = \frac{2\pi}{3}$$

.

Therefore, the angle between **u** and **v** is $2\pi/3$ (or $120°$).

Exercise 12 Using dot products

What is the angle between these each pair of vectors:

- $[1, 0]$, $[0, 1]$
- $[3, 4]$, $[4, 3]$
- $[2, -1, 2]$, $[-1, 2, -2]$
- $[-5, 0, -1]$, $[2, 3, -4]$

Working Space

Answer on Page 101

4.3 Dot products in Python

NumPy will let you do dot products using the symbol @. Open `first_vectors.py` and add the following to the end of the script:

```
# Take the dot product
    d = v @ u
    print("v @ u =", d)

    # Get the angle between the vectors
    a = np.arccos(d / (mv * mu))
    print(f"The angle between u and v is {a * 180 / np.pi:.2f} degrees")
```

When you run it you should get:

```
v @ u = 4
The angle between u and v is 78.55 degrees
```

4.4 Work and Power

Earlier, we mentioned that mechanical work is the product of the force you apply to something and the amount it moves (formulaic representation: $W = \vec{F} \cdot \vec{d}$). For example, if you push a train with a force of 10 newtons as it moves 5 meters, you have done 50 joules of work.

What if you try to push the train sideways? It moves down the track 5 meters, but you push it as if you were trying to derail — perpendicular to its the tracks and its motion. You have done no work, because the train didn't move at all in the direction you were pushing.

Now that you know about dot products: The *work* you do is the dot product of the force vector you apply and the displacement vector of the train. (The displacement vector is the vector that tells how the train moved while you pushed it.)

$$W = \vec{F} \cdot \vec{d} = |\vec{F}||\vec{d}| \cos\theta$$

$$P = \vec{F} \cdot \vec{v} = |\vec{F}||\vec{v}| \cos\theta$$

Similarly, we mentioned that *power* is the product of the force you apply and the velocity of the mass you are applying it to. It is actually the dot product of the force vector and the velocity vector.

For example, if you are pushing a sled with a force of 10 newtons and it is moving 2 meters per second, but your push is 20 degrees off, you aren't transferring 20 watts of power to the sled. You are transferring $10 \times 2 \times \cos(20 \text{ degrees}) \approx 18.8$ watts of power.

Exercise 13 Power at an angle

A person pushes a skateboard with a force of 40 N. The skateboard is moving forward at 3 m/s, but the person's push is angled 30° above the direction of motion. How much power is being transferred to the skateboard?

Working Space

Answer on Page 101

Similarly, the same thing happens when pushing vertically, against forces such as gravity. If a cart moves a distance L up a ramp at angle θ, its displacement is $\vec{d} = [L\cos\theta, L\sin\theta]$. Gravity is $\vec{F}_g = [0, -mg]$.

Therefore,

$$\vec{F}_g \cdot \vec{d} = [0, -mg] \cdot [L\cos\theta, L\sin\theta] = -mgL\sin\theta.$$

Since $L \sin \theta$ is the height of the ramp, gravity does $-mgh$ joules of work. This is where the get that a change in potential energy is a form of work being done.

Exercise 14 Work done on a Box

A force $\vec{F} = [12, 5]$ N pushes a box whose displacement is $\vec{d} = [3, 0]$ m. How much work is done?

Working Space

Answer on Page 102

Chapter 5

Manufacturing

If you try to think of any man-made object, whether it was made from woods, metals, or plastics, chances are it was produced through a manufacturing process.

Over time, these processes have been refined to be more efficient, cost-effective, and faster at producing the goods that we use on a daily basis.

New methods are also constantly being developed by engineers and scientists, and today the range of options available means that choosing the most appropriate manufacturing method involves finding the sweet spot between cost-effectiveness, yield, and time needed.

5.1 Woods and Metals Processes

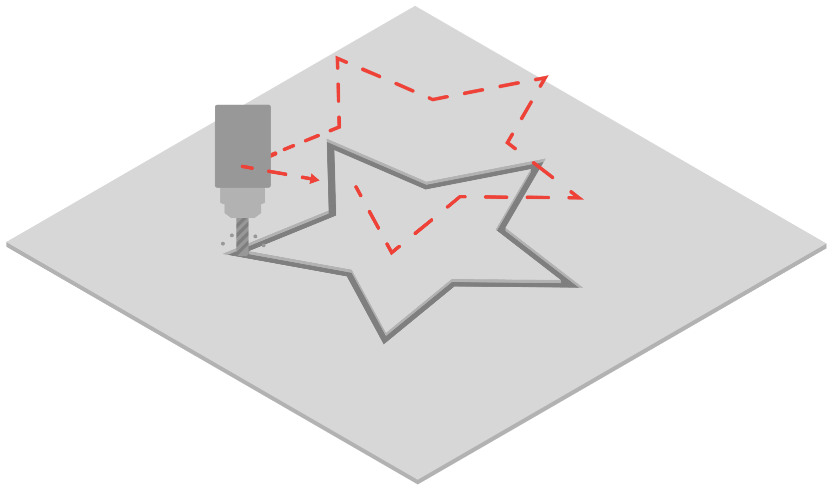
Manufacturing methods that are able to process woods and metals are typically the processes that are used to construct the vast majority of the built world around us.

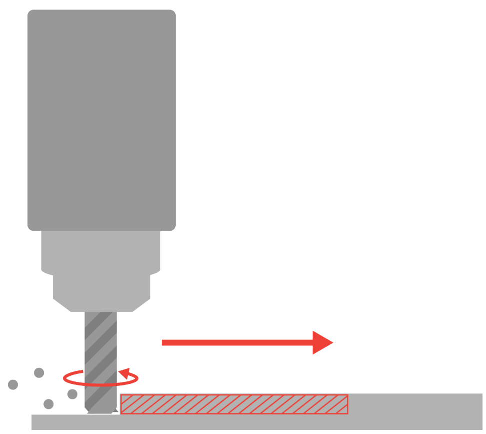
Infrastructure, transportation methods, and buildings would not exist without the advent of processes that allow us to accurately machine raw woods and metals into our desired forms.

The following sections will cover methods used to process both material types.

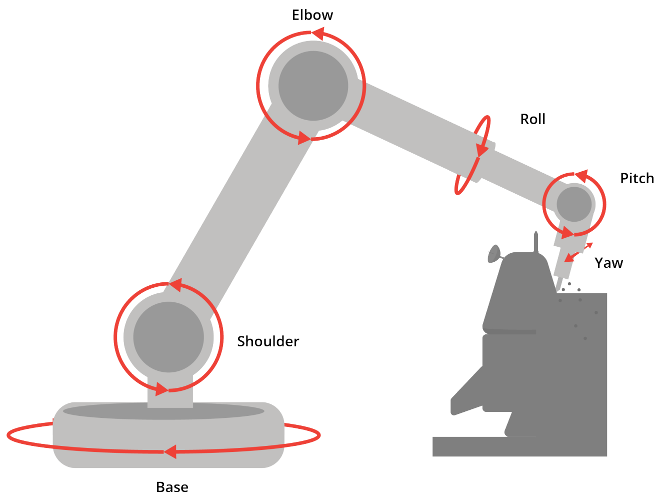
5.1.1 Milling

Mills are powerful tools that allow us to carve out complex shapes from blocks of raw material. A tool bit follows a path to remove the desired material, which makes it a *subtractive* manufacturing process. The tool bit rotates at a very high speed, which allows it to process harder materials such as woods and metals.





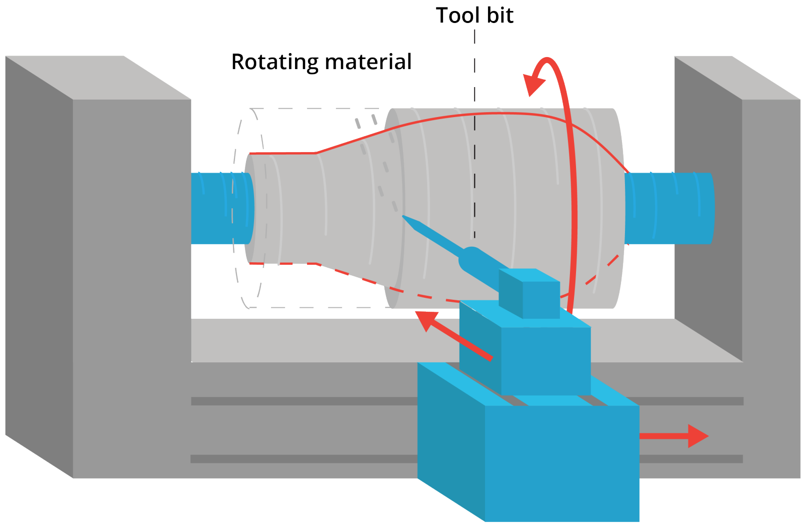
There are various types of mills, ranging from a 3-axis mill, which can cut out simple shapes in the X, Y, and Z axes, all the way to 6-axis mills, which can also rotate about those axes to create more complex curvatures.



In manufacturing settings where speed and repeatability is paramount, mills are often computer controlled. This functionality is referred to as *Computer Numerical Control*, or CNC. CNC mills are able to repeatedly follow a tool path, resulting in consistent and accurate parts.

5.1.2 Lathing

Lathes are tools that allow us to carve out complex cylindrical shapes from raw material. Like a mill, it also is a subtractive manufacturing process. However, lathes rotate the material itself at a high speed, rather than the tool bit. As the material rotates, the tool bit can be used to extract material layer by layer.



In the manufacturing industry, lathes are also often computer controlled. Alongside CNC mills, these CNC lathes are responsible for a majority of the objects that we interact with everyday.

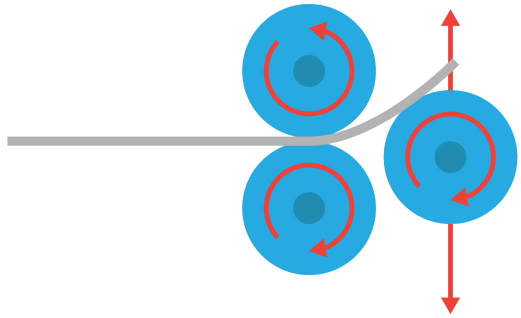
5.2 Metal-specific Processes

While mills and lathes are already able to cover the vast majority of manufacturing needs for woods and metals, there are certain processes that are specifically enabled by the unique properties of metals. More often than not, these processes leverage the malleability (ability to bend without breaking) of metals at room temperature or higher.

5.2.1 Sheet Metal Processes

While milling allows us to process blocks of metal to great effect, sometimes the application we need does not require material of such thickness.

This is where sheet metal comes in, as well as the methods we use to process it. One of the most common techniques in manufacturing is rolling, where a raw sheet is gradually rolled into the desired shape. This method is used to create many objects you may recognize, such as metal roofings, aircraft frames, and more.

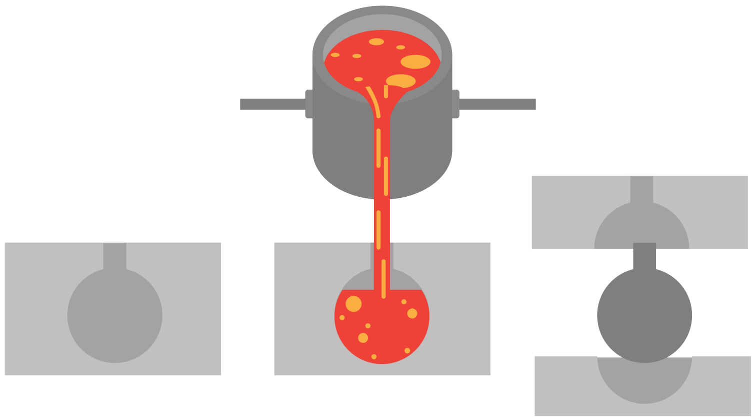


Another method is stamping, where a raw sheet is stamped into the desired shape. This allows us to create surfaces with complex geometries in an instant, and in large quantities. This method is often used for applications like the exterior panels of a car, where parts with compound curves are needed.

Lastly, there are also separate processes used to increase the strength of sheet metal parts. This falls under the category of sheet metal forming, and involves bending the edges of a part to form a reinforced flap. Almost all sheet metal parts are reinforced in this manner, as it is a relatively simple process and also helps to create a clean edge for a more finished look.

5.2.2 Casting

The last kind of metal-specific process we will cover is casting. Casting involves pouring molten metal into a mold, then letting the metal cool and set inside the mold. Smaller components with complex geometries and limited structural requirements (such as toys) are often cast, as it is an accurate and high-volume manufacturing method.



Casting also results in minimal material wastage, as it is not a subtractive manufacturing method where excess material is machined away, but rather only the specific amount of material required is poured in each time.

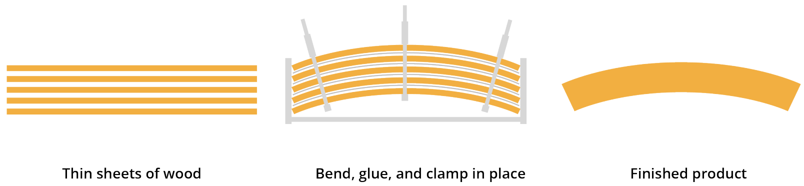
5.3 Wood-specific Processes

Similar to metals, there are also certain manufacturing processes that are enabled by the unique qualities of wood. These make use of the water content inherent in wood, and the flexibility it enables.

5.3.1 Bending

Turning raw wood into flat, workable pieces involves a variety of tools that you probably know of already, such as saws and drills. However, there are specific processes that help us create curved shapes with wood, in addition to the mill and the lathe mentioned earlier.

This is where bent lamination comes in. Bent lamination involves layering multiple thin veneers or strips of wood with adhesive, and clamping it to create the desired form while letting the glue dry. This method is often used for furniture production, enabling continuous curves in wood with tight radii.



Steam can also be used for bending, by helping soften the wood fibers to increase flexibility. Once the desired form is reached, the part can then cool down and harden. Unlike bent lamination, steam bending can be done without adhesives.

5.4 Plastic-specific Processes

Although plastics only came into prominence in the mid 20th century, they have changed manufacturing and, by extension, the world as we know it. Easily manufacturable, durable, and cost-effective, they have come to permeate almost everything we use on a daily basis.

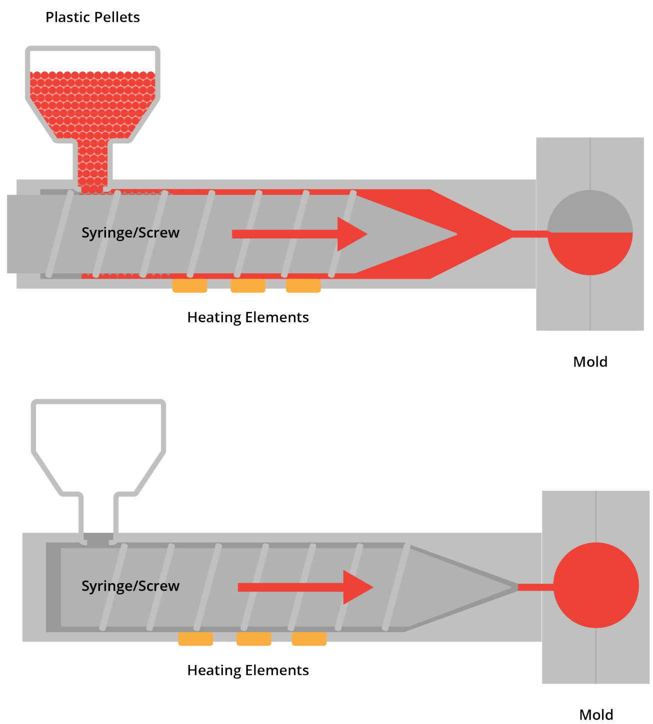
It must also be noted that these exact qualities have also resulted in the proliferation of plastics in our environment, and as such, usage of plastics should be well considered and limited. Alternative biodegradable materials are currently being trialed by scientists and would look to replicate many of the same qualities we expect from plastics, including its manufacturability.

5.4.1 Injection Molding

Injection molding is responsible for the vast majority of plastic products that you interact with on a daily basis. It is extremely quick, highly accurate, and has minimal material wastage, making it a popular and cost effective method of manufacturing plastic goods.

Similar to casting, injection molding involves injecting molten plastic into a mold, then allowing the part to cool and set inside the cavity.

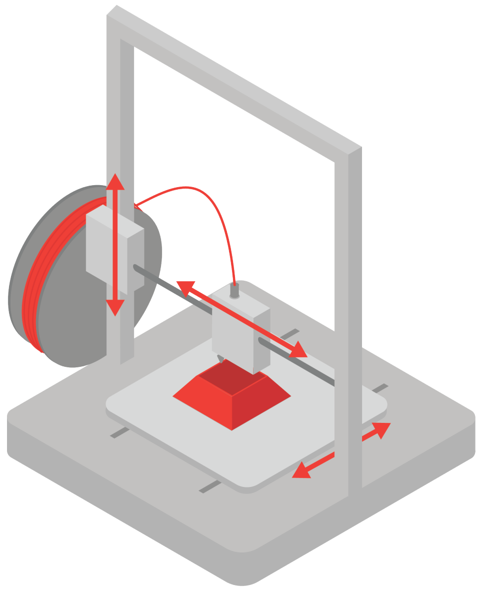
You can often tell when a part was produced through injection molding, with telltale signs such as the parting line. This is where the parts of the mold meet, forming a visible line on the surface of a part.



5.5 3D Printing

Injection molding has traditionally been the go-to technique for manufacturing plastic goods. However, new technologies result in the advent of new manufacturing methods. 3D printing is one such method, having come to prominence in the last few decades as a way to quickly prototype parts without having to create the molds needed for injection molding.

3D printing is an *additive* manufacturing process, where molten material is applied layer by layer to form the desired geometry. It allows for complex geometries, standardized batch production, and whilst the accuracy may currently lag behind traditional injection molding, it is also improving rapidly.

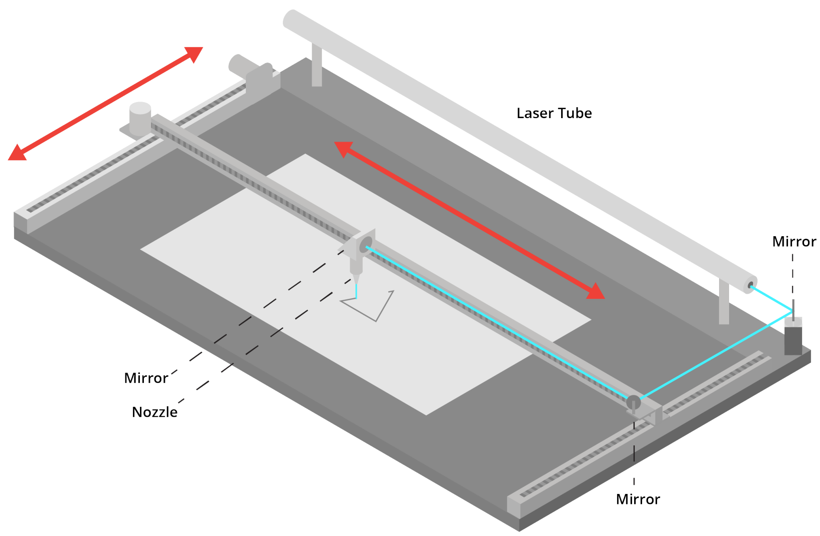


5.6 Laser Cutting and Water Jet

Similarly, another manufacturing method enabled by new technologies is laser cutting. Like 3D printing, it has come into prominence as a method to quickly prototype parts. However, it is not an additive manufacturing process.

Instead laser cutting uses a laser beam to cut and etch through sheets of plastic, however it can also be used for other materials such as fabrics and card stocks. Laser cutting is mostly limited by material thickness, and as such can only cut through thinner sheets of material.

Similarly, water jetting uses pressurized water to blast a stream of water through material (metal, wood, stone, or rubber usually). As the nozzle moves, the high pressure water traces a path throughout the material. Water jets are also limited to material thickness, as too thick of material may not easily be cut.


Laser Tube
Mirror
Mirror
Nozzle
Mirror

Chapter 6

Introduction to Waves

When we talk about waves, we refer to so many phenomena in nature. Waves can be in the form of liquids, like when you throw a pebble into a pond and create a recurring effect of water. Waves can be ocean waves, controlled by the moon and the tides. Waves can be sound waves, bouncing back and forth travelling from a speaker. Waves can even my magnetic or electric, which we call electromagnetic waves, creating charge and magnetic fields. Light waves creates the large range of frequency of light we see with the human eye. All of these waves, however, have something in common: *they all share the same wave properties.*

We define waves as follows:

> A **wave** is a disturbance that transfers energy from one place to another without permanently transferring matter.

Another way to put it:

> A **wave** is a vibration of a system out of a restful state.

For example, when a water wave moves across a pond, the water itself does not travel across the pond. Instead, the water particles oscillate around their equilibrium positions while the wave carries energy outward.

In the case of a spring, which we will analyze deeper, energy (by the form of stretching or pushing) is transferred throughout the medium that is the spring.

6.1 Basic Properties of Waves

All waves can be described using several important properties. You will definitely see an overlap between trigonometric functions and wave properties. Often, we can describe waves that are generated with these trigonometric functions we have previously talked about.

There are essentially 5 terms we use to describe the properties of waves: **equilibrium**, **amplitude**, **wavelength**, **period**, and **frequency**.

Equilibrium

An **equilibrium** state of a wave is what it would look no oscillation to be present. In other words, it is a state of rest for the wave. If you have a body of water that a pebble is thrown into, the equilibrium state is height or state of the water the instant before the pebble is thrown in. Same for a rope, a equilibrium state of a rope represents the rope before any motion is put into it.

Amplitude

When a wave is created, it moves creates motion away from its *equilibrium* position.

The **amplitude** of a wave is the maximum displacement of the medium from its equilibrium position. In simpler terms, it measures how “large” the wave is.

The distance between the highest positive amplitude and the equilibrium position (called the *peak* or *crest*) and the lowest positive amplitude and the equilibrium position (called the *valley* or *trough*) are the same distance.

For a water wave, the amplitude is the height from the resting water level to the crest of the wave. For a sound wave, greater amplitude corresponds to louder sound. In light waves, larger amplitude corresponds to greater brightness.

Wavelength

The **wavelength**, usually represented by the symbol λ (lambda), is the distance between two identical points on consecutive waves, such as crest-to-crest or trough-to-trough.

Wavelength is usually measured in meters (or centimeters or micrometers).

Frequency

The **frequency** of a wave is the number of complete waves that pass a point each second. Frequency is measured in hertz (Hz), where

$$1\ \text{Hz} = 1\ \text{wave per second}.$$

A wave with a higher frequency oscillates more rapidly. For sound waves, higher frequency corresponds to higher pitch. For light waves, frequency determines the color of the light.

Period

The **period** of a wave is the time required for one complete oscillation. The period and frequency are related by

where:

- T is the period (seconds),
- f is the frequency (hertz).

Wave Speed

Waves travel at a certain speed depending on the medium through which they move. Wave speed is related to frequency and wavelength by

$$v = f\lambda$$

where:

- v is wave speed,
- f is frequency,
- λ is wavelength.

ADD: exercise in labelling

6.2 Two main types of waves

In waves around us, we can classify them under two categories: **longitudinal** and **transverse**. This classification depends on how particles are effected as the wave "passes".

Longitudinal

For *longitudinal* waves, the direction of wave (or the distrubance in particles) is *parallel* to the direction of the vibration. The most common example of this a *sound wave*. A speaker playing a sound pushes one sound particle, pushing another, and then another, ultimately continuing until the particles reach your ears. There is an amazing visual of this using a slinky in your digital resources!

Transverse

In contrast, *transverse* waves move in a direction (or the distrubance in particles) is *perpendicular* to the wave's direction. Light and Electromagnetic waves and light waves are examples of transverse waves. The energy moves left to right (or horizontally) but the ends of the wave slide up and down vertically.

Again, see your digital resources for a visual of this using a slinky.

6.3 Compression and Rarefaction

Sound waves travel by creating disturbances in a medium such as air. When an object vibrates, it pushes nearby air particles together in some regions and spreads them farther apart in others. These alternating regions move outward through the medium as the sound wave travels.

Regions where particles are crowded closer together are called *compressions*. In these regions, the pressure and density of the air are higher. Regions where particles are spread farther apart are called *rarefactions*[1]. In these regions, the pressure and density are lower.

As the wave passes, the particles themselves do not travel long distances with the wave. Instead, they oscillate back and forth around their equilibrium positions while the disturbance moves through the medium.

6.4 Creating a Wave in Python

ADD ME

[1] Rarefaction regions and phases are commonly called decompression regions.

Chapter 7

Sound

When you set off a firecracker, it makes a sound.

Let's break that down a little more. Inside the cardboard wrapper of the firecracker, there is potassium nitrate (KNO_3), sulfur (S), and carbon(C). These are all solids. When you trigger the chemical reactions with a little heat, these atoms rearrange themselves to be potassium carbonate (K_2CO_3), potassium sulfate (K_2SO_4), carbon dioxide (CO_2), and nitrogen (N_2). Note that the last two are gasses.

The molecules of a solid are much more tightly packed than the molecules of a gas. So after the chemical reaction, the molecules expand to fill a much bigger volume. The air molecules nearby get pushed away from the firecracker. They compress the molecules beyond them, and those compress the molecules beyond them.

This compression wave radiates out as a sphere; its radius growing at about 343 meters per second ("The speed of sound").

The energy of the explosion is distributed around the surface of this sphere. As the radius increases, the energy is spread more and more thinly around. This is why the firecracker seems louder when you are closer to it. (If you set off a firecracker in a sewer pipe, the sound will travel much, much farther.)

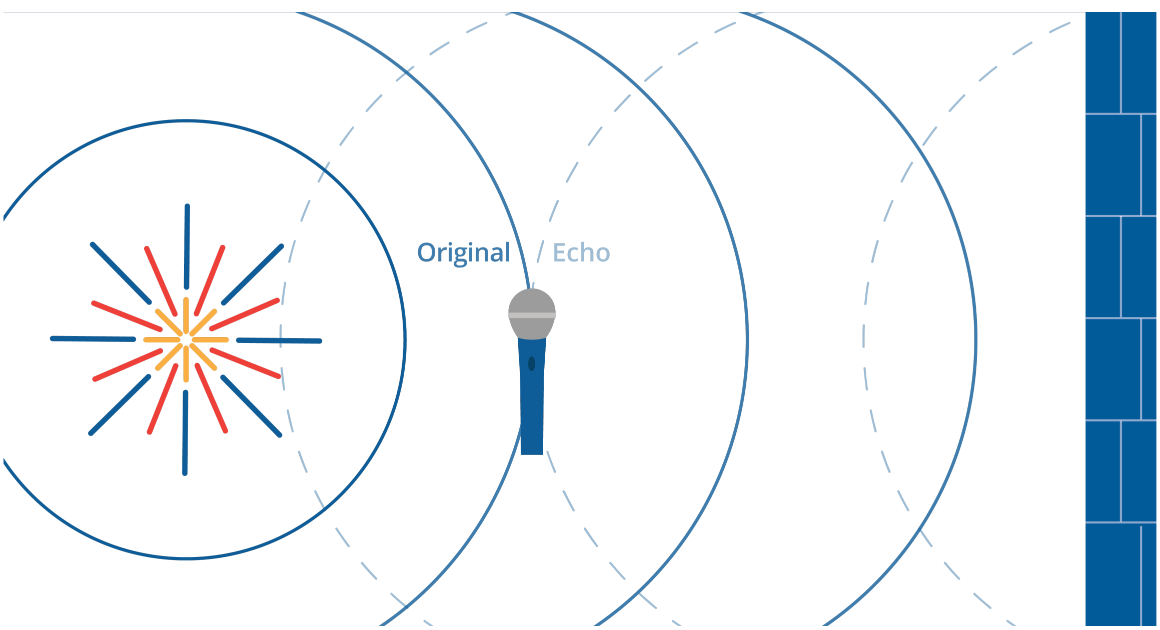


Figure 7.1: A firecracker exploding causes an initial sound wave and an echo.

This compression wave will bounce off of hard surfaces. If you set off a firecracker 50 meters from a big wall, you will hear the explosion twice. We call the second one an "echo".

The compression wave will be absorbed by soft surfaces. If you covered that wall with pillows, there would be almost no echo.

The study of how these compression waves move and bounce is called *acoustics*. Before you build a concert hall, you hire an acoustician to look at your plans and tell you how to make it sound better.

7.1 Pitch and frequency

The string on a guitar is very similar to the weighted spring example. The farther the string is displaced, the more force it feels pushing it back to equilibrium (remember the tension force?). Thus, it moves back and forth in a sine wave. (OK, it isn't a pure sine wave, but we will get to that later.)

The string is connected to the center of the boxy part of the guitar, which is pushed and pulled by the string. That creates compression waves in the air around it.

If you are in the room with the guitar, those compression waves enter your ear and push and pull your eardrum, which is attached to bones that move a fluid that tickles tiny hairs, called *cilia*, in your inner ear. This is how you hear.

We sometimes see plots of sound waveforms. The x-axis represents time. The y-axis represents the amount the air is compressed at the microphone that converted the air pressure into an electrical signal.

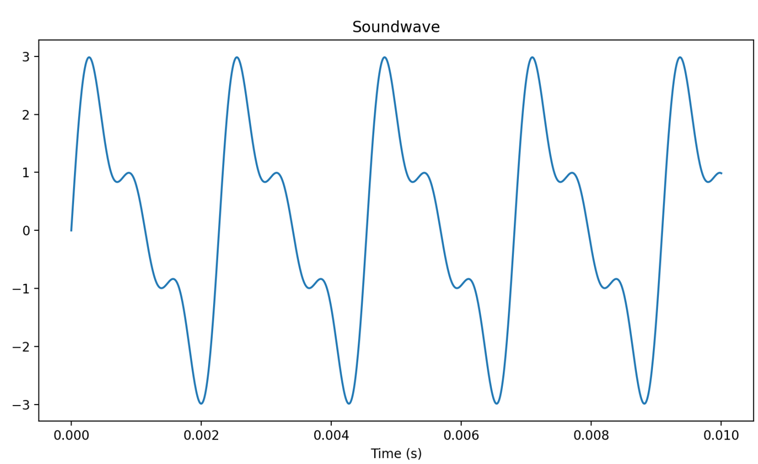


Figure 7.2: A soundwave graph over time.

If the guitar string is made tighter (by the tuning pegs) or shorter (by the guitarist's fingers on the strings), the string vibrates more times per second. We measure the number of waves per second and we call it the *frequency* of the tone. The unit for frequency is *Hertz*: cycles per second. The *period* is opposite of frequency: it is the time it takes for one cycle to complete. The unit for period is seconds per cycle.

Musicians have given the different frequencies names. If the guitarist plucks the lowest note on his guitar, it will vibrate at 82.4 Hertz. The guitarist will say "That pitch is low E." If the string is made half as long (by a finger on the 12th fret), the frequency will be twice as fast (164.8 Hertz), and the guitarist will say "That is E an octave up."

For any note, the note that has twice the frequency is one octave up. The note that has half the frequency is one octave down.

The octave is a very big jump in pitch, so musicians break it up into 12 smaller steps. If the guitarist shortens the E string by one fret, the frequency will be $82.4 \times 1.059463 \approx 87.3$ Hertz.

Shortening the string one fret always increases the frequency by a factor of 1.059463. Why?

Because $1.059463^{1}2 = 2$. That is, if you take 12 of these hops, you end up an octave higher. This, the smallest hop in western music, is referred to as a *half step*.

Exercise 15 Notes and frequencies

Working Space

The note A near the middle of the piano, is 440Hz. The note E is 7 half steps above A. What is its frequency?

Answer on Page 102

7.2 Chords and harmonics

Of course, a guitarist seldom plays only one string at a time. Instead, they use the frets to pick a pitch for each string and strums all six strings.

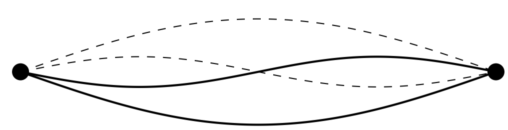
Some combinations of frequencies sound better than others. We have already talked about the octave: If one string vibrates twice for each vibration of another, they sound sweet together.

Musicians speak of "the fifth". If one string vibrates three times and the other vibrates twice in the same amount of time, they sound sweet together.

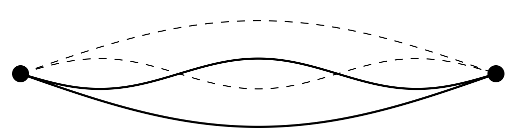
Likewise, if one string vibrates 4 times while the other vibrates 3 times, they sound sweet together. Musicians call this "the third."

Each of these different frequencies tickle different cilia in the inner ear, so you can hear all six notes at the same time when the guitarist strums their guitar.

When a string vibrates, it doesn't create a single sine wave. Yes, the string vibrates from end-to-end, and this generates a sine wave at what we call *the fundamental frequency*. However, there are also "standing waves" on the string. One of these standing waves is still at the centerpoint of the string, but everything to the left of the centerpoint is going up, while everything to the right is going down. This creates *an overtone* that is twice the frequency of the fundamental.



The next overtone has two still points — it divides the string into three parts. The outer parts are up, while the inner part is down. Its frequency is three times the fundamental frequency.



And so on. 4 times the fundamental, 5 times the fundamental, etc.

In general, tones with many overtones tend to sound bright. Tones with just the fundamental sound thin.

Humans can generally hear frequencies from 20Hz to 20,000Hz (or 20kHz). Young people tend to be able to hear very high sounds better than older people.

Dogs can generally hear sounds in the 65Hz to 45kHz range.

7.3 Making waves in Python

Let's make a sine wave and add some overtones to it. Create a file named `harmonics.py`.

```
import matplotlib.pyplot as plt
import math

# Constants: frequency and amplitude
fundamental_freq = 440.0 # A = 440 Hz
fundamental_amp = 2.0

# Up an octave
first_freq = fundamental_freq * 2.0 # Hz
first_amp = fundamental_amp * 0.5

# Up a fifth more
second_freq = fundamental_freq * 3.0 # Hz
second_amp = fundamental_amp * 0.4

# How much time to show
max_time = 0.0092 # seconds

# Calculate the values 10,000 times per second
time_step = 0.00001 # seconds

# Initialize
time = 0.0
times = []
totals = []
fundamentals = []
firsts = []
seconds = []

while time <= max_time:
    # Store the time
    times.append(time)

    # Compute value each harmonic
    fundamental = fundamental_amp * math.sin(2.0 * math.pi * fundamental_freq *
    ↪ time)
    first = first_amp * math.sin(2.0 * math.pi * first_freq * time)
    second = second_amp * math.sin(2.0 * math.pi * second_freq * time)

    # Sum them up
    total = fundamental + first + second

    # Store the values
    fundamentals.append(fundamental)
    firsts.append(first)
    seconds.append(second)
    totals.append(total)
```

```

    # Increment time
    time += time_step

# Plot the data
fig, ax = plt.subplots(2, 1)

# Show each component
ax[0].plot(times, fundamentals)
ax[0].plot(times, firsts)
ax[0].plot(times, seconds)
ax[0].legend()

# Show the totals
ax[1].plot(times, totals)
ax[1].set_xlabel("Time (s)")

plt.show()
```

When you run it, you should see a plot of all three sine waves and another plot of their sum:

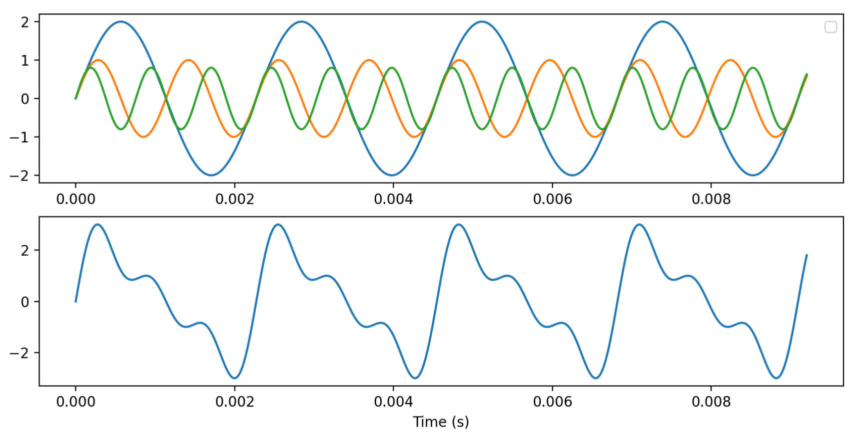


Figure 7.3: The output of haronics.py

7.3.1 Making a sound file

The graph is pretty to look at, but make let's a file that we can listen to.

The WAV audio file format is supported on pretty much any device, and a library for writing WAV files comes with Python. Let's write some sine waves and some noise into a WAV file.

Create a file called `soundmaker.py`

```
import wave
import math
import random

# Constants
frame_rate = 16000 # samples per second
duration_per = 0.3 # seconds per sound
frequencies = [220, 440, 880, 392] # Hz
amplitudes = [20, 125]
baseline = 127 # Values will be between 0 and 255, so 127 is the baseline
samples_per = int(frame_rate * duration_per) # number of samples per sound

# Open a file
wave_writer = wave.open('sound.wav', 'wb')

# Not stereo, just one channel
wave_writer.setnchannels(1)

# 1 byte audio means everything is in the range 0 to 255
wave_writer.setsampwidth(1)

# Set the frame rate
wave_writer.setframerate(frame_rate)

# Loop over the amplitudes and frequencies
for amplitude in amplitudes:
    for frequency in frequencies:
        time = 0.0
        # Write a sine wave
        for sample in range(samples_per):
            s = baseline + int(amplitude * math.sin(2.0 * math.pi * frequency * time))
            wave_writer.writeframes(bytes([s]))
            time += 1.0 / frame_rate

        # Write some noise after each sine wave
        for sample in range(samples_per):
            s = baseline + random.randint(0, 15)
            wave_writer.writeframes(bytes([s]))

# Close the file
wave_writer.close()
```

When you run it, it should create a sound file with several tones of different frequencies and volumes. Each tone should be followed by some noise.

Chapter 8

Applying Numerical Methods to Bivariate Data

8.1 Review of Scatter Plots

8.2 Correlation Coefficient

In the previous section, we used scatterplots to visually judge whether two variables are related. But how do we move beyond ”this looks like a strong relationship” to something we can actually calculate and compare? That is exactly what the *correlation coefficient* gives us.

A correlation coefficient is a numerical measure used to judge the relation between two variables. The one we will use throughout this course is *Pearson's correlation coefficient*, also known simply as the correlation coefficient. It measures the strength and direction of the linear relation between two quantitative variables.

You will encounter two versions of this value depending on whether you are working with a full population or a sample drawn from one. When computed from a population, the correlation coefficient is written as ρ (read as “rho”). When computed from a sample, it is written as r. In this course, we will almost always be working with sample data, so r is the symbol you will use most.

No matter how strong or weak a relationship is, r always falls within a fixed range:

$$-1 \leq r \leq +1$$

Think about what those endpoints mean. A value of $r = +1$ or $r = -1$ means the points in a scatterplot fall perfectly on a straight line. A value of $r = 0$ means there is no linear relationship between the two variables at all. Every other value of r falls somewhere in between, and in the next subsection we will look at exactly how to interpret where r lands.

Exercise 16 Boundaries of r

Answer the following questions about the range of the correlation coefficient.

1. A student computes $r = 1.3$ for a dataset. What does this tell you, without looking at the data?
2. Two researchers study the same population but one uses the full population and the other uses a random sample. What symbol does each researcher use for their correlation coefficient?
3. Can r equal exactly 0.5? Can it equal exactly -1? Explain what each of those values would mean about the data.

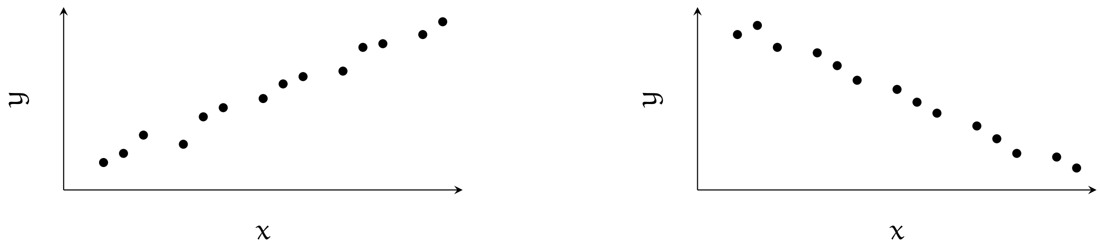
Working Space

Answer on Page 102

Direction

Now that we know r always falls between -1 and $+1$, the first thing to read from it is direction. The sign of r tells you immediately whether the two variables move together or against each other.

A positive value of r indicates that as x increases, y also tends to increase. Taller fathers tend to have taller sons, so the scatterplot shows an upward trend and r is positive.A negative value of r indicates that as x increases, y tends to decrease. As a phone gets older, its battery life tends to decrease, so the scatterplot shows a downward trend and r is negative.



(a) Positive relation ($r > 0$) (b) Negative relation ($r < 0$)

Figure 8.1: Direction of a linear relation as shown in a scatterplot.

Notice that direction says nothing about strength. Both $r = +0.1$ and $r = +0.95$ are positive, but they describe very different relationships. That is what we look at next.

Exercise 17 Reading Direction from r

Working Space

For each pair of variables, predict whether r would be positive, negative, or zero.

x	y	Sign of r
Hours studied	Exam score	
Temperature	Hot drinks sold	
Shoe size	Intelligence	
Age of a car	Resale value	

Answer on Page 103

Strength

Direction tells us which way a relationship goes, but not how strong it is. For that, we look at the numeric value of r, ignoring its sign. The further r is from zero, the stronger the linear relationship between the two variables. The closer r is to zero, the weaker it is.

At the extremes, $r = +1$ or $r = -1$ indicates a *perfect correlation*, meaning every point in the scatterplot falls exactly on a straight line. In practice, real data almost never achieves this. On the other end, $r = 0$ means there is no linear relationship at all.

To compare the strength of two correlations, just compare the absolute values. If $r = -0.86$ for one pair of variables and $r = 0.75$ for another, the first pair has the stronger linear relationship because $0.86 > 0.75$.

Since interpreting strength can be subjective, statisticians often use cutoff values as a rough guide. The diagram below shows one commonly used set of cutoffs. Keep in mind that these are conventions, not rules. The key principle is always the same: the further from zero, the stronger the correlation.

Very Strong	Strong	Weak	No Correlation	Weak	Strong	Very Strong
−1.00 to −0.85	−0.85 to −0.50	−0.50 to −0.10	−0.10 to +0.10	+0.10 to +0.50	+0.50 to +0.85	+0.85 to +1.00

Figure 8.2: One commonly used set of cutoff values for interpreting the strength of a correlation coefficient.

Exercise 18 Interpreting Strength

Working Space

The table below shows correlation coefficients computed from four different datasets. For each one, describe the direction and approximate strength of the relationship.

r	Direction	Strength
+0.92		
−0.34		
+0.07		
−0.88		

Answer on Page 103

Example

Let us put direction and strength together using a real dataset. Consider a sample of 10 father-son pairs. The heights (in inches) are recorded below.

Height of Father x (inches)	Height of Son y (inches)
66	66
66	65
67	67
67	70
68	67
68	70
69	70
70	72
71	72
73	74

Before computing anything, we check whether a correlation coefficient is appropriate. Correlation only applies to linear relationships, so we first make a scatterplot to confirm the relation between the two variables is linear.

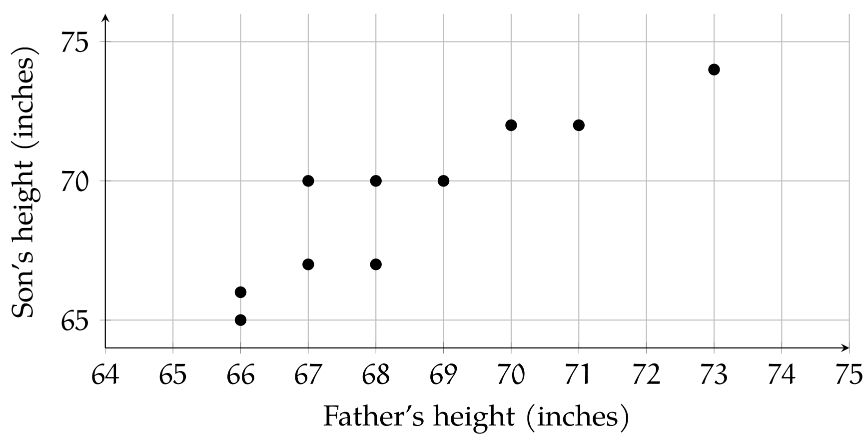


Figure 8.3: Scatterplot of father and son heights for a sample of 12 pairs.

The scatterplot shows a linear trend, so it is appropriate to compute r. In practice, correlation coefficients are computed using a calculator or statistical software rather than by hand. For this dataset, the computed value is approximately $r = 0.96$.

Computing r in Excel

Open your data in Excel or Google Sheets with the x values in column A and the y values in column B. To compute r directly, use the built-in correlation function:

```
=CORREL(A2:A11, B2:B11)
```

This returns the correlation coefficient between the two columns instantly. For the father-son data, this should return approximately 0.96.

Computing r in Python

```
import numpy as np

# Father and son heights in inches
x = [66, 66, 67, 67, 68, 68, 69, 70, 71, 73]
y = [66, 65, 67, 70, 67, 70, 70, 72, 72, 74]

r = np.corrcoef(x, y)[0, 1]
print(r)
```

`np.corrcoef()` returns a 2×2 matrix of correlation coefficients. The value at position `[0, 1]` is the correlation between x and y. For this dataset, the output should be approximately `0.96`.

Try replacing the data with your own values and observe how r changes as the relationship between the two variables becomes stronger or weaker.

What does this tell us? The sign is positive, so taller fathers tend to have taller sons. The value is 0.96, which falls in the very strong range of our cutoff table. The points cluster tightly around a straight line, and the scatterplot confirms this.

Notice also what r does not tell us. It describes the strength and direction of the linear relationship, but not the shape of any curve, and not whether one variable causes the other. A strong correlation between two variables does not mean one causes the other to change.

Exercise 19 Interpreting a Computed r

Working Space

Answer on Page 103

sectionLinear Regression

Suppose you wanted to predict a teacher's starting salary based on how many years of experience they have. You might expect that more experience leads to a higher salary, but how do you turn that intuition into a precise estimate? That is exactly what a *linear regression model* does. It gives you a straight-line equation that describes the relationship between two variables, so you can use the value of one to predict the value of the other.

The Regression Model

When one variable depends on or is explained by another, we call the variable being predicted the *dependent variable*. The variable doing the explaining is the *independent variable*. In our example, starting salary is the response variable and years of experience is the independent variable, it would make little sense to say that a teacher's salary determines how many years they have worked.

The linear relation between two variables is described by the *population regression equation*:

$$Y = \alpha + \beta X$$

where Y is the response variable, X is the independent variable, α is the y-*intercept* (the value of Y when $X = 0$), and β is the *slope* (the amount Y changes for every one-unit increase in X). This equation describes the true relationship in the population, but in practice, we never know α and β exactly. We estimate them from sample data.

Once we have estimates for α and β, we can write the *estimated regression line*:

$$\hat{y} = a + bx$$

The symbol $\hat{y}$ (read "y-hat") is the *predicted value* of Y for a given value of X. Notice the difference in notation: α and β are population parameters; a and b are their sample-based estimates.

Now, no data will ever fall perfectly on a straight line. A student who studies ten hours might score differently than the line predicts. The difference between what the model predicts and what is actually observed is called the *residual*, or *error*, and is denoted e:

$$e = y - \hat{y}$$

A positive residual means the observed value was higher than predicted; a negative residual means it was lower. Going back to our teachers: if the model predicts a starting salary of \$38,000 for a teacher with four years of experience, but that teacher actually earns \$40,500, the residual is $40{,}500 - 38{,}000 = 2{,}500$. This teacher earned \$2,500 more than the model expected.

Exercise 20 Computing a Residual

A regression model predicts that a teacher with seven years of experience will have a starting salary of $42,300. The teacher's actual starting salary is $40,100.

1. Compute the residual.
2. Was the teacher's salary over-predicted or under-predicted? Explain.

Working Space

Answer on Page 103

The goal of linear regression is to find the line that makes these residuals as small as possible across all observations.

The Least-Squares Regression Line

Every straight line you could draw through a scatterplot will produce a different set of residuals. Some lines will over-predict for certain point and under-predict for others. We want to know: which line does the best job overall? The answer is the *least-squares regression line*, also called the *line of best fit*. It is the line that minimizes the sum of the squared residuals, so, it makes $\sum e^2 = \sum(y - \hat{y})^2$ as small as possible.

Why should we square the residuals? First, squaring eliminates the sign, so positive and negative errors do not cancel each other out. Second, squaring penalizes large errors more heavily than small ones, which pushes the line toward a better overall fit. Now you know!

Exercise 21 Why Square the Residuals?

Suppose a regression line produces the following residuals for five data points: $3, -3, 2, -1, -1$.

1. Compute the sum of the residuals. What do you notice?
2. Compute the sum of the squared residuals.
3. Explain why simply minimizing the sum of the residuals would not be a useful goal.

Working Space

Answer on Page 103

The least-squares regression line has two properties that always hold, regardless of the dataset.

First, it always passes through the point $(\bar{X}, \bar{Y})$ which represent the means (or average points) of the two variables. This point is sometimes called the *centroid* of the data.

Second, the slope b of the line is always given by:

$$b = r\left(\frac{S_y}{S_x}\right)$$

where r is the correlation coefficient, S_y is the standard deviation of the Y-values, and S_x is the standard deviation of the X-values. Notice that the slope depends directly on r: if there is no linear relationship ($r = 0$), the slope is zero, and the best prediction for any X is simply $\bar{Y}$.

Once you have b, the y-intercept a follows from the fact that the line passes through $(\bar{X}, \bar{Y})$:

$$a = \bar{Y} - b\bar{X}$$

These two formulas together give you the equation of the least-squares regression line:

$$\hat{y} = a + bx$$

Exercise 22 Computing the Line

For a dataset of teachers' years of experience (X) and starting salary in thousands of dollars (Y), the following summary statistics were computed: $\bar{X} = 6$, $\bar{Y} = 44$, $S_x = 3.2$, $S_y = 5.8$, $r = 0.91$.

1. Compute the slope b of the least-squares regression line.
2. Compute the y-intercept a.
3. Write the equation of the least-squares regression line.
4. The line must pass through the centroid of the data. Verify this using your equation.

Working Space

Answer on Page 104

Interpreting Slope and Intercept

Having the equation of the least-squares regression line is only half the job. The other half is being able to explain what it means in plain language.

The slope b tells you how much the predicted value of Y changes for every one-unit increase in X. The key phrase you could see on an exam is **on average** because b is an estimate based on sample data, not an exact rule.

For our teachers example, if $b = 1.65$, the correct interpretation is:

> For every additional year of experience, the predicted starting salary increases by $1,650, on average.

Notice three things about that sentence. It names both variables. It gives the direction and amount of change. And it includes "on average." Leave out any one of those and the interpretation is incomplete.

Exercise 23 Interpreting the Slope

A least-squares regression line for predicting a teacher's starting salary (in thousands of dollars) from years of experience has slope $b = 1.49$.

1. Write a correct interpretation of the slope in context.
2. A student writes: "For every extra year of experience, salary goes up by $1,490." What is wrong with this interpretation?

Working Space

Answer on Page 104

The y-intercept a is the predicted value of Y when $X = 0$. Interpreting it requires care. Sometimes $X = 0$ is meaningful where a teacher with zero years of experience is simply a newly hired teacher. In that case the intercept has a natural interpretation: it is the predicted starting salary for someone entering the profession with no prior experience.

Other times $X = 0$ is outside the range of the data entirely, or physically impossible. In those situations, the intercept is mathematically necessary to define the line but has no useful real-world meaning. You should note that and move on.

Exercise 24 Interpreting the Intercept

Using the regression equation $\hat{y} = 34.10 + 1.649x$, where x is years of experience and $\hat{y}$ is predicted starting salary in thousands of dollars:

1. Write a correct interpretation of the y-intercept in context.
2. Is this interpretation meaningful here? Explain.

Working Space

Answer on Page 104

Worked Example

A random sample of 10 teachers was selected from a large urban school district. Each teacher's years of experience at the time of hiring and their starting salary (in thousands of dollars) were recorded. The data is given in the table below.

Experience (years)	Starting Salary ($1000s)
3	36
8	43
1	32
0	30
5	38
10	47
2	33
7	42
4	37
6	41

Before doing any calculations, look at the data. As years of experience increase, starting salaries appear to rise steadily. There are no obvious outliers, and the values are spread across a reasonable range of experience levels. A scatterplot would confirm whether this relationship is linear and this is! There is a strong positive linear relationship between experience and starting salary.

From the data, we have $n = 10$ pairs of measurements. Using a calculator or software, we can obtain the following summary statistics:

$$\bar{X} = 4.6, \quad \bar{Y} = 37.9, \quad S_x = 3.169, \quad S_y = 5.013, \quad r = 0.9964$$

Exercise 25 Slope, Intercept, and the Regression Equation

Using the summary statistics above:

1. Compute the slope b of the least-squares regression line. Interpret it in context.
2. Compute the y-intercept a. Interpret it in context.
3. Write the equation of the least-squares regression line.

Working Space

Answer on Page 104

Exercise 26 Prediction, Residuals, and R^2

Using the regression equation $\hat{y} = 30.650 + 1.576x$:

1. Predict the starting salary for a teacher with 5 years of experience.
2. The teacher in the sample with 8 years of experience has an actual starting salary of $43,000. Compute the residual and interpret it.
3. Compute the coefficient of determination R^2 and interpret it in context.
4. About what percentage of the variation in starting salaries is *not* explained by years of experience? Suggest one factor that might account for this.

Working Space

Answer on Page 105

textbfComputing in Excel

You do not need to compute the slope, intercept, and R^2 by hand every time. Excel can produce all of these values at once, along with a scatterplot showing the line of best fit. Here is how.

First, enter your data into two columns, with experience values in column A and starting salary values in column B, with headers in row 1. Select both columns, then insert a scatterplot using **Insert** $\rightarrow$ **Chart** $\rightarrow$ **Scatter**. Once the chart appears, click on any data point, then select **Add Trendline**. In the trendline options panel on the right, choose **Linear** and check both **Display Equation on chart** and **Display R-squared value on chart**. Excel will draw the line of best fit and print the equation and R^2 directly on the chart.

To get the full regression output, use the **Data Analysis** Toolpak. Go to **Data** $\rightarrow$ **Data Analysis** $\rightarrow$ **Regression**. Set the Input Y Range to your salary column and the Input X Range to your experience column. Check **Labels** if your columns have headers. Click OK and Excel will produce a full regression table in a new sheet. The values you need are in the **Coefficients** column: the row labelled **Intercept** gives you a, and the row labelled

Experience gives you b. The R^2 value appears near the top of the output under **R Square**.

Computing in Python

Python can produce the same regression results as Excel, and a little more besides. The code below uses two libraries: `numpy` for the summary statistics and `scipy` for the regression, and `matplotlib` to draw the scatterplot with the line of best fit.

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

# Data
experience = [3, 8, 1, 0, 5, 10, 2, 7, 4, 6]
salary     = [36, 43, 32, 30, 38, 47, 33, 42, 37, 41]

# Least-squares regression
slope, intercept, r, p, se = stats.linregress(experience, salary)

print(f"Slope (b):     {slope:.3f}")
print(f"Intercept (a): {intercept:.3f}")
print(f"r:             {r:.4f}")
print(f"R-squared:     {r**2:.4f}")

# Scatterplot with regression line
x_line = np.linspace(0, 10, 100)
y_line = intercept + slope * x_line

plt.scatter(experience, salary, color="steelblue", label="Data")
plt.plot(x_line, y_line, color="firebrick",
         label=f"y = {intercept:.2f} + {slope:.2f}x")
plt.xlabel("Experience (years)")
plt.ylabel("Starting Salary ($1000s)")
plt.title("Experience vs. Starting Salary")
plt.legend()
plt.show()
```

Running this code produces the following output:

```
Slope (b):     1.673
Intercept (a): 30.203
r:             0.9957
R-squared:     0.9915
```

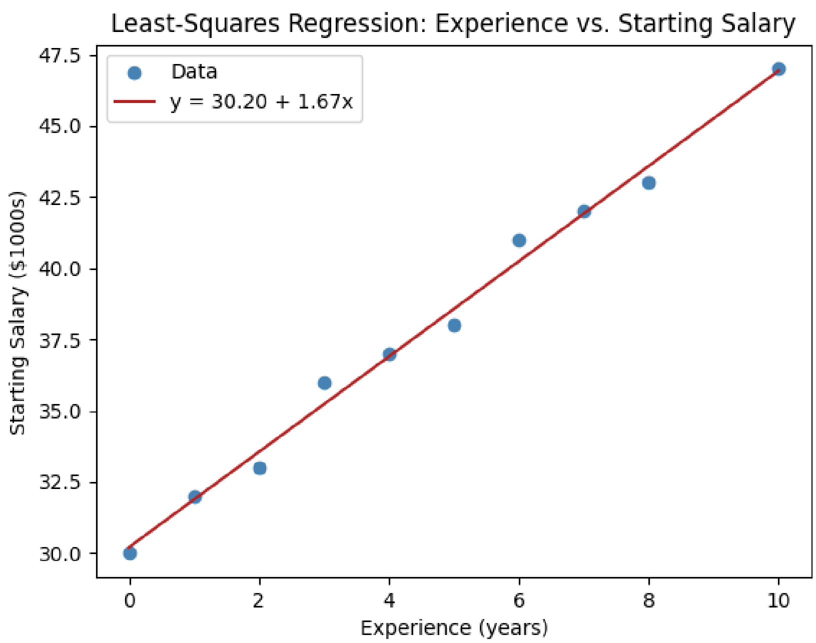


Figure 8.4: Experience vs. Starting Salary Graph generated from Python

These match the values we computed by hand exactly. The `stats.linregress` function does all the heavy lifting! It returns the slope, intercept, correlation coefficient r, a p-value, and the standard error of the slope in one line. For now, focus on the first four outputs. The p-value and standard error will become important when we study inference for regression later!

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 23)

1. $\sum_{i=1}^{7} X_i = 22 + 19 + 25 + 21 + 18 + 23 + 20 = 148.$
2. $(2) + (-1) + (5) + (1) + (-2) + (3) + (0) = 8$. The sum decreased by $7 \times 20 = 140$, which is n times the constant subtracted. Shifting every value by a constant shifts the total by n times that constant.

Answer to Exercise 2 (on page 26)

1. $\bar{X} = \frac{12 + 15 + 9 + 11 + 14 + 8 + 13 + 52}{8} = \frac{134}{8} = 16.75$ books.
2. Sorted: 8, 9, 11, 12, 13, 14, 15, 52. $l = \frac{9}{2} = 4.5$, so $M = \frac{12 + 13}{2} = 12.5$ books.
3. The mean is more affected. It uses every value in the calculation, so the outlier of 52 pulls it upward. The median depends only on the middle two values, which are unaffected by how extreme the highest value is.
4. Without 52: $\bar{X} = \frac{82}{7} \approx 11.71$ books. Sorted: 8, 9, 11, 12, 13, 14, 15; median $= 12$. The mean dropped by about 5 books; the median changed by only 0.5.

Answer to Exercise 3 (on page 27)

1. Mean. Symmetric data with no outliers means the mean and median are close, and the mean is preferred for further calculations.
2. Median. The CEO's salary is an extreme outlier that would pull the mean well above what a typical employee earns. The median better represents the center of the bulk of the data.
3. Median. The long right tail indicates right skew; the mean would be pulled toward

the older ages and away from where most people cluster.

4. Mean. Once the outlier is removed and the distribution is symmetric, the mean is the appropriate choice.

Answer to Exercise 4 (on page 31)

1. $s^2 = \frac{800.38}{14-1} = \frac{800.38}{13} \approx 61.57$ cars^2.

   Note: the exact sum of squared deviations is ≈ 742.21, giving $s^2 \approx 57.09$. If your computation gives a slightly different value due to rounding $\bar{X}$, that is expected. Using the unrounded mean gives $s \approx 7.56$ cars.

2. $s = \sqrt{57.09} \approx 7.56$ cars.

3. s is measured in cars. s^2 is measured in cars^2, which has no natural interpretation, which is one reason we prefer the standard deviation over the variance for reporting.

4. The second dealership. Both have the same mean, but a smaller standard deviation means the daily sales are more tightly clustered around the average.

Answer to Exercise 5 (on page 33)

1. Standard deviation. Symmetric data with no outliers is exactly the situation where the mean and standard deviation are appropriate.

2. IQR. The extreme high values would inflate the standard deviation substantially, making it a misleading measure of typical spread. IQR focuses on the middle 50% and is unaffected by the luxury properties.

3. Standard deviation. Once the distribution is symmetric and outlier-free, the standard deviation is the preferred measure and will now accurately reflect the spread of the remaining data.

Answer to Exercise 6 (on page 36)

Multiplying every value by $b = 1.02$:

Measure	Original	After multiplying by 1.02
Mean	12,000	$1.02(12{,}000) = 12{,}240$
Median	8,000	$1.02(8{,}000) = 8{,}160$
Range	30,000	$1.02(30{,}000) = 30{,}600$
Std dev	5,000	$1.02(5{,}000) = 5{,}100$
Q_1	5,000	$1.02(5{,}000) = 5{,}100$
Q_3	14,000	$1.02(14{,}000) = 14{,}280$
IQR	9,000	$1.02(9{,}000) = 9{,}180$

Unlike adding a constant, multiplying affects the spread measures too. Every distance in the distribution is stretched by the same factor.

Answer to Exercise 7 (on page 39)

1. $z = \dfrac{24 - 14.64}{7.56} \approx 1.24$. The best day was about 1.24 standard deviations above the mean. It was a notably good day but not extreme.
2. $z = \dfrac{2 - 14.64}{7.56} \approx -1.67$. The worst day was about 1.67 standard deviations below the mean and further from typical than the best day was above it.
3. Second dealership: $z = \dfrac{31 - 20}{4} = 2.75$. The second dealership's best day was more unusually high at 2.75 standard deviations above its mean, compared to 1.24 for the first dealership.
4. Still $z \approx 1.24$. Adding a constant to every value shifts the mean by the same constant and leaves the standard deviation unchanged. The relative position of every observation in the distribution stays the same, so z-scores are unaffected.

Answer to Exercise 8 (on page 42)

1. $l = \dfrac{(14+1)(50)}{100} = 7.5$. The 50th percentile falls halfway between the 7th observation (14) and the 8th (17): $P_{50} = \dfrac{14+17}{2} = 15.5$. This matches the median exactly.
2. $Q_3 = P_{75} = 21$, so 75% of days had 21 or fewer cars sold.
3. $Q_1 = 9$ and $Q_3 = 21$. The value 17 lies between $Q_2 = 15.5$ and $Q_3 = 21$, placing it in the third quarter (between the median and Q_3). The consultant is correct.
4. Days with 2 or fewer cars: only the day with 2 cars sold. That is 1 out of 14 days, which is about 7% consistent with the bottom 10% cutoff of 2.5.

Answer to Exercise 9 (on page 45)

The nodes in this graph are unique words. The edges represent a valid one letter operations that can transform one word into another word. These operations include adding a letter, removing a letter, or changing a letter into another. For example, the word `cat` would be connected to `cot` (change), `cart` (add), and `cut` (change), but not to `dog`. Usually this will be represented as a tree, which is a special type of graph we will cover later.

Here is an example of a word graph:

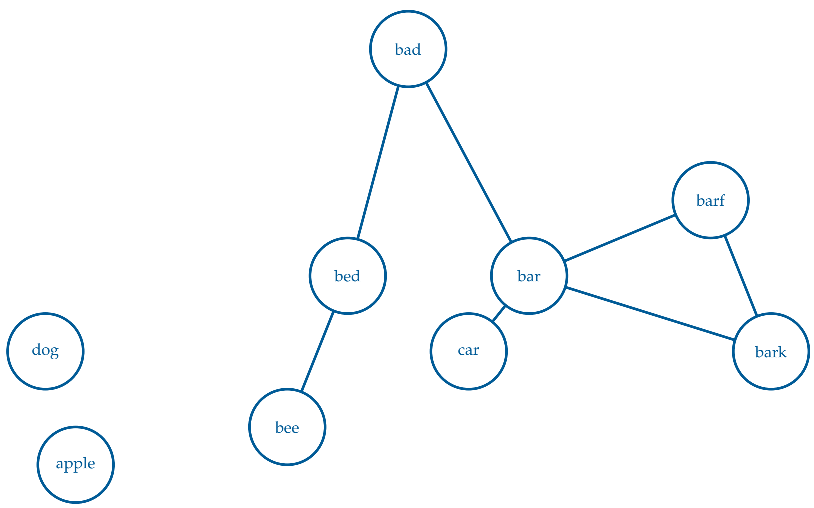


Figure 3.4: A word graph where the nodes are unique words and the edges represent a one letter operations between two words.

Answer to Exercise 10 (on page 47)

In a weighted graph, each node would have to store a set of neighbors and their respective weights. The best way to do this would be a tuple. For example,

- A: [(B, 4), (C, 3)]
- B: [(A, 4), (D, 8)]
- C: [(A, 3)]
- D: [(B, 8)]

Answer to Exercise 11 (on page 51)

- $[1, 2, 3] \cdot [4, 5, -6] = 4 + 10 - 18 = -4$
- $[\pi, 2\pi] \cdot [2, -1] = 2\pi - 2\pi = 0$
- $[0, 0, 0, 0] \cdot [10, 10, 10, 10] = 0 + 0 + 0 + 0 = 0$

Answer to Exercise 12 (on page 56)

- $[1, 0] \cdot [0, 1] = 0$. The angle must be $\pi/2$.
- $[3, 4] \cdot [4, 3] = 24$. $|[3, 4]||[4, 3]| \cos(\theta) = 24$. $\cos(\theta) = \frac{24}{(5)(5)}$. $\theta = \arccos(\frac{24}{25}) \approx 0.284$ radians.
- $[2, -1, 2] \cdot [-1, 2, -2] = 4-2-4 = -2$. $|[2, -1, 2]| = \sqrt{4+1+4} = \sqrt{9} = 3$. $|[-1, 2, -2]| = \sqrt{1+4+4} = \sqrt{9} = 3$. $3(3)\cos\theta = -2$. $\theta = \arccos(-2/9) \approx 1.795$ radians.
- $[-5, 0, -1] \cdot [2, 3, -4] = -10+0+4 = -6$. $|[-5, 0, 1]| = \sqrt{25+0+1} = \sqrt{26}$. $|[2, 3, -4]| = \sqrt{4+9+16} = \sqrt{29}$. $\sqrt{26}(\sqrt{29})\cos\theta = -6$. $\theta = \arccos(\frac{-6}{\sqrt{26}\sqrt{29}}) \approx 1.791$ radians.

Answer to Exercise 13 (on page 57)

Power is the dot product of the force vector and the velocity vector:

$$P = \vec{F} \cdot \vec{v}$$

Since the force and velocity are at an angle, we can use:

$$P = |\vec{F}||\vec{v}| \cos\theta$$

Here, $|\vec{F}| = 40$ N, $|\vec{v}| = 3$ m/s, and $\theta = 30°$:

$$P = 40(3)\cos(30°)$$

Since $\cos(30°) \approx 0.866$:

$$P \approx 120(0.866) \approx 104 \text{ W}$$

Only the part of the force pointing in the direction of motion transfers power to the skateboard. The angled part still matters physically, but it does not all help move the skateboard forward. So the power transferred to the skateboard is about 104 watts.

Answer to Exercise ?? (on page 58)

The work done is the dot product of the force vector and the displacement vector:

$$W = \vec{F} \cdot \vec{d}$$

Here, the force is $\vec{F} = [12, 5]$ N and the displacement is $\vec{d} = [3, 0]$ m. So, we use vector multiplication but multiplying each component:

$$W = [12, 5] \cdot [3, 0] = 12(3) + 5(0) = 36 \text{ J}$$

The horizontal part of the force does work because the box moves horizontally. The vertical part of the force does no work because the box does not move vertically. So the total work done is 36 joules.

Answer to Exercise 15 (on page 75)

A is 440 Hz. Each half-step is a multiplication by $\sqrt[12]{2} = 1.059463094359295$ So the frequency of E is $(440)(2^{7/12}) = 659.255113825739859$

Answer to Exercise 16 (on page 82)

1. A value of $r = 1.3$ is impossible. The correlation coefficient must satisfy $-1 \leq r \leq +1$, so the student made a calculation error.
2. The researcher using the full population uses ρ. The researcher using a sample uses r.
3. Both are possible. An r of 0.5 would indicate a moderate positive linear relationship. The points follow an upward trend but with noticeable scatter. An r of -1 would indicate a perfect negative linear relationship, meaning all points fall exactly on a downward-sloping line.

Answer to Exercise 17 (on page 83)

x	y	Sign of r
Hours studied	Exam score	Positive
Temperature	Hot drinks sold	Negative
Shoe size	Intelligence	Zero
Age of a car	Resale value	Negative

Answer to Exercise 18 (on page 84)

r	Direction	Strength
$+0.92$	Positive	Very strong
-0.34	Negative	Weak
$+0.07$	Positive	No correlation
-0.88	Negative	Very strong

Answer to Exercise 19 (on page 86)

Answer to Exercise 20 (on page 88)

1. $e = 40{,}100 - 42{,}300 = -2{,}200$.
2. The salary was over-predicted. The observed salary was \$2,200 less than the model expected.

Answer to Exercise 21 (on page 89)

1. $3 + (-3) + 2 + (-1) + (-1) = 0$. The positive and negative residuals cancel out, making the sum zero even though the line does not fit perfectly.
2. $3^2 + (-3)^2 + 2^2 + (-1)^2 + (-1)^2 = 9 + 9 + 4 + 1 + 1 = 24$.
3. A line that over-predicts some values and under-predicts others by equal amounts would have a residual sum of zero but it would be a poor fit. Squaring forces every error to contribute positively, so the minimization is meaningful!

Answer to Exercise 22 (on page 90)

1. $b = 0.91\left(\frac{5.8}{3.2}\right) = 0.91 \times 1.8125 \approx 1.649$
2. $a = 44 - 1.649(6) = 44 - 9.897 \approx 34.103$
3. $\hat{y} = 34.103 + 1.649x$
4. Substituting $x = \bar{X} = 6$: $\hat{y} = 34.103 + 1.649(6) = 34.103 + 9.897 = 44.000 = \bar{Y}$. ✓

Answer to Exercise 23 (on page 91)

1. For every additional year of teaching experience, the predicted starting salary increases by $1,490, on average.
2. The interpretation is missing the phrase "on average" and the word "predicted." The slope is an estimate from sample data, not a guarantee that every teacher's salary increases by exactly $1,490 per year.

Answer to Exercise 24 (on page 92)

1. The predicted starting salary for a teacher with zero years of experience is $34,100.
2. Yes, it is meaningful in this context. A teacher with no prior experience is a realistic case, so $X = 0$ is within the plausible range of the data.

Answer to Exercise 25 (on page 93)

1. $$b = r\left(\frac{S_y}{S_x}\right) = 0.9964\left(\frac{5.013}{3.169}\right) \approx 1.576$$
   For every additional year of teaching experience, the predicted starting salary increases by $1,576, on average.
2. $$a = \bar{Y} - b\bar{X} = 37.9 - 1.576(4.6) \approx 30.650$$
   The predicted starting salary for a teacher with no prior experience is $30,650. Since

$X = 0$ represents a newly hired teacher with no experience, this interpretation is meaningful in context.

3. The equation of the least-squares regression line is:

$$\hat{y} = 30.650 + 1.576x$$

where x is years of experience and $\hat{y}$ is the predicted starting salary in thousands of dollars.

Answer to Exercise 26 (on page 94)

1.

$$\hat{y} = 30.650 + 1.576(5) = 30.650 + 7.880 = 38.530$$

The predicted starting salary for a teacher with 5 years of experience is $38,530.

2.

$$\hat{y} = 30.650 + 1.576(8) = 30.650 + 12.608 = 43.258$$

$$e = y - \hat{y} = 43 - 43.258 = -0.258$$

The residual is −$258. The teacher's actual starting salary was $258 less than the model predicted. The salary was slightly over-predicted.

3.

$$R^2 = r^2 = (0.9964)^2 \approx 0.9928$$

About 99.28% of the variation in starting salaries is explained by the linear relationship with years of experience. This is an exceptionally strong fit.

4. About $100 - 99.28 = 0.72\%$ of the variation is unexplained. This is very small, but it may be due to factors such as the teacher's level of education, the subject area taught, or the specific school within the district.

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